

Throughput-Optimal Multiresource-Job Scheduling with Continuous Requirement Distribution

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Abstract

Modern computing systems process jobs with resource requirements such as CPU and memory, which are described by multiresource jobs (MRJ) queueing models. In practice, job resource requirements are spread out over so many values, that it is rare to see the same value twice. This pattern is best modeled by a continuous distribution of requirement values. However, the existing theoretical work on stability or throughput-optimality focuses on queueing models with class-based resource requirements. In class-based models, the number of distinct resource requirements must be small to demonstrate strong empirical performance, making them a poor match for these practical systems.

We introduce the first throughput-optimal family of scheduling policies for the continuous MRJ model, with both preemptive and nonpreemptive variants. We further introduce several efficient policy families, which remain throughput-optimal while considerably improving computational efficiency, under some distributional assumptions. We use a discretization approach, where we choose the discretization granularity based on the system load and the distribution of resource requirements. We validate the real-world applicability of our policies by comparing them against existing index-based policies on parametrized distributions and on datacenter trace data from the Google Borg scheduler, demonstrating state-of-the-art performance.

Keywords: Multiresource-Job, Scheduling, Queueing, Continuous Resource Requirement, Discretization-Based Policies, Throughput-Optimality

1 Introduction

Modern computing systems, such as cloud platforms, datacenters, and personal machines, typically possess multiple kinds of computational resources (e.g., CPU, GPU, memory). These systems can be modeled as queues where jobs require certain amounts of these resources, in order to be served. We refer to these jobs as multiresource jobs.¹ The system is able to serve any set of multiresource jobs simultaneously, provided that the total requirement of each resource does not exceed the corresponding resource capacity.

A scheduling policy specifies which set of multiresource jobs should be served at any time. Since the 1980s, many policies have been extensively studied [1–13], and their analyses include strong results such as system stability and throughput-optimality in their respective settings.

However, existing policies, such as MaxWeight [5, 6], Randomized Multi-Resource Scheduling (RMS) [14], Markovian Service Rate (MSR) [15], are restricted to settings with a finite set of possible resource requirements, which are known as “job classes”. For instance, MaxWeight weights jobs by how many jobs of identical resource requirements are present. The larger the number of job classes, the less the empirical performance corresponds to the theoretical results. In particular, the number of jobs present must be of at least the order of the number of job classes for empirical performance to correspond to the theoretical results. These queue lengths are too large for practical purposes. In datacenter scheduling traces, hundreds of thousands of unique resource requirements are commonly seen [16]. Theoretical results, which only kick in above hundreds of thousands jobs, are not useful.

In real-world scheduling environments, for example, the first million jobs of the Borg trace [16] have 3,668 unique memory requirements, 536,229 unique CPU requirements, and 949,217 unique (CPU, memory) pairs. As a result, the existing class-based policies used in the literature are not suitable for such practical settings. Seeing this mismatch, we introduce the *continuous MRJ model*, in which the distribution of the resource requirement is continuous. While real systems do not literally have continuous resource requirements, the continuous MRJ model captures the behavior of rarely repeating resource requirements. In this model, we ask a question:

Does there exist a throughput-optimal family of scheduling policies in the continuous MRJ setting?

We resolve this question by developing the first throughput-optimal family of scheduling policies, the $\vec{K}$ -discretized MaxWeight ($\vec{K}$ -MW) family. $\vec{K}$ -MW uses discretization intentionally, rather than assuming that the resource requirements are inherently well-discretized. Our proof builds off the existing class-based MaxWeight

¹When the system is equipped with only one resource type, we refer to these jobs as multiserver jobs.

proof, with additional techniques to handle the overhead of discretization, which must be controlled and mitigated.

Many applications involve nonpreemptible jobs or require substantial preemption overhead. We therefore introduce the first nonpreemptive throughput-optimal family of scheduling policies: $\vec{K}$ -discretized nonpreemptive Markovian Service Rate ($\vec{K}$ -nMSR). For $\vec{K}$ -nMSR, we use the same discretization approach as $\vec{K}$ -MW and construct the policy using the offline precomputation method for the MSR policies introduced in Chen et al. [15].

In addition, computational efficiency is essential for both the online optimization in MaxWeight-type policies and the offline precomputation in nMSR-type policies. For instance, when using standard class-based MaxWeight in systems with numerous classes and small expected service durations, the exponentially large number of possible service options may cause the system to spend more time updating service options than actually serving jobs. To achieve computational efficiency, we must avoid handling an extremely large candidate set. We introduce a new group of policies that restrict attention to smaller candidate sets, where we only consider a small fraction of the possible service options, during MaxWeight and nMSR’s optimization steps. We call these smaller candidate sets “efficient sets”.

We start by considering only the service options that form extreme vertices of the convex hull of the set of all service options. By doing so, for example, we reduce the candidate set by around $200\times$ when $K = 60$ in the single-resource setting. Next, in the settings with symmetrical resource requirement distribution, our 2-Job efficient set reduces the number of service options to a quantity proportional to the number of discretization classes. In the single-resource setting with decreasing requirement density, our 2-Bucket efficient set likewise considers a number of service options proportional to the number of discretization classes. Our policies built on these three efficient sets are each throughput-optimal under corresponding distributional assumptions on the resource requirements.

We use stimulation to extensively verify the performance, both throughput and mean response time, in both parameterized settings and a public Borg trace. Our simulation results verify the throughput optimality of the K -MW family and the Efficient K -MW families on a variety of requirement distributions. Moreover, both on these distributions and in experiments based on job requirements extracted from the Google Borg trace, our theoretically-motivated policies outperform in mean response time existing index-based policies: FCFS, First-Fit, Best-Fit, and Least-Server-First (LSF). Note that LSF is equivalent to naive MaxWeight in the continuous MSJ setting, because every job in its own class has equal weight. Naive MaxWeight then always serves most jobs, and packs the jobs in increasing order of their resource requirement.

This paper is organized as follows:

- Section 2 reviews related prior work on multiserver and multiresource scheduling.
- Section 3 introduces the multiresource model under the continuous resource requirement setting, the $\vec{K}$ -discretization scheme, and our proposed policies.
- Section 4 presents our main theorems, including throughput-optimality results for each family of policies.
- Section 5 provides detailed proofs of the main theorems.

- Section 6 compares our policies with existing policies via simulations.

2 Prior Work

In this section, we review real-world scheduling practices in datacenters and supercomputers, as well as the theoretical development of scheduling policies for multiresource-job systems.

2.1 Scheduling in Large-Scale Computing Settings

Traditional one-server-per-job queueing systems are not suitable for modeling many modern computing environments. A more accurate representation is provided by the multiserver-job (MSJ) system, where a job requires some amount of a single resource, and the multiresource-job (MRJ) system, where a job requires some amount of multiple resources. LLM handling infrastructure, such as within Microsoft Azure [17], handles inference tasks, which vary by having a different number of tokens in different requests, corresponding to an MSJ model. The number of tokens implicitly corresponds to GPU and memory requirements.

Similarly, traditional datacenters process jobs that use varying amounts of resources in multiple categories, such as CPU, memory, and more, corresponding to an MRJ model. These multiresource requirements are present at the Google Borg [16, 18–23] and Google Omega [24] systems, as well as Alibaba’s Sigma and Fuxi systems [25–27]. Finally, high-performance computing jobs submitted to supercomputers require a pre-specified number of nodes, for instance, at Argonne’s Aurora [28] and Oak Ridge’s Frontier [29, 30] supercomputers, corresponding to an MSJ model.

Furthermore, the distribution of these resource requirements has extremely large support, subdivided smaller than any useful measure of precision. For instance, the traces available from Google Borg [20, 21, 31] demonstrate that CPU and memory requests are specified as floats with at least 13 digits of precision, and have nearly no repeated pairs: around 95% unique pairs over the first 1 million requirements. The RAM usage in Google Omega is recorded in bytes and is therefore specified down to the individual byte, e.g., 1074000000, 82587) [24]. In Microsoft Azure, although token requirements are discretely represented, traces from multiple LLM inference services over a single day include 2339 distinct context-token counts and 623 distinct generated-token counts [17]. In all of these scenarios, class-based scheduling, where the “class” refers to a set of jobs with identical resource requirements, becomes unsuitable, as two jobs with only minor differences in some resource requirements would be assigned to distinct classes. This observation motivates the need to consider scheduling policies under the settings where resource requirements are continuous or of large support.

2.2 Multiserver-job and Multiresource-job Scheduling

We now review prior work on scheduling policies for the MSJ and MRJ settings. Some policies are designed primarily for the MSJ system and do not extend to the MRJ system.

The policies proposed in prior work can be broadly divided into two categories: index-based policies and class-based policies. Specifically, a job class refers to a set of jobs with identical resource requirements. Class-based policies schedule based on the number of jobs in each class that are present in the system at a given time. In other words, class-based policies rely on job-type information when making scheduling decisions. This implicitly assumes that job classes are discrete and finite, and that the number of possible classes is smaller than the number of jobs typically present in the system, so that the queue-length information for each class is meaningful. In contrast, index-based policies do not classify jobs into finitely many types. Instead, an index-based policy orders the jobs in the system according to a certain rule, iterates over all jobs or a subset of jobs, and decides whether to serve each job in turn. Table 1 summarizes the properties of some representative index-based and class-based policies in the prior work compared with our policies introduced in Section 3. Notably, all prior theoretical works prove analytical results only in settings where resource requirements are finitely supported. To the best of our knowledge, no theoretical results have been established for continuous-resource settings, which are the focus of this paper.

Policies	Nonpreemptive	MRJ Setting	Throughput-Optimal	Continuous Resource
FCFS	✓	✓	✗	Analysis only in discrete setting
First-Fit	✓	✓	✓ under assumptions [4]	Analysis only in discrete setting
Best-Fit	✓	✓	✓ under assumptions [4]	Analysis only in discrete setting
LSF	✓	✓	✗	Analysis only in discrete setting
ServerFilling	✓	✗	✓ under assumptions [9]	Analysis only in discrete setting
ServerFilling-SRPT	✗	✗	✓ under assumptions [10]	Analysis only in discrete setting
MaxWeight	✗	✓	✓	✗
RMS	✓	✓	✓	✗
MSR	✓	✓	✓	✗
MSFQ	✓	✗	✓	✗
$\vec{K}$ -MW	✗	✓	✓	✓
$\vec{K}$ -MSR	✓	✓	✓	✓

Table 1: Summary of the settings and the throughput-optimal condition of existing index-based and class-based policies, and our $\vec{K}$ -discretized MaxWeight ($\vec{K}$ -MW) and $\vec{K}$ -discretized nonpreemptive Markovian Service Rate ($\vec{K}$ -nMSR) families of policies.

We now introduce some classical index-based policies. As a representative index-based policy, First-Come-First-Served (FCFS) has been studied since the 1980s. Stability and mean response time analyses in the two-server setting were conducted in [1, 2]. In the integer-valued requirement setting, the stability region of FCFS was characterized in [7], and the first mean response time analysis was conducted in [11].

Other well-known index-based policies include First-Fit, Best-Fit, and Least-Server-First (LSF), each of which uses Backfilling to greedily pack jobs according to some scanning order. We elaborate on the definitions of these three policies in Appendix B. First-Fit and Best-Fit are throughput-optimal if the requirement distribution is symmetric [4]. However, they generally fail to be throughput-optimal in more general settings [6]. Hong and Wang [12] studied the mean response time of LSF in a low-load scaling regime, showing that LSF achieves the same asymptotic growth rate as the optimal mean response time bound. However, LSF is generally not throughput-optimal because it prioritizes small jobs and fails to fully utilize system capacity. ServerFilling [9] and ServerFilling-SRPT [10] are recently introduced near-index policies, with strong mean response bounds in the power-of-two MSJ requirement setting. Although most of these index-based policies can naturally be applied to the continuous MSJ or MRJ framework, their theoretical analysis has focused only on discrete settings, where job requirements have finite support. Moreover, none of them is throughput-optimal in the general resource requirement distribution setting.

We next move to some well-known class-based policies. One of the earliest preemptive class-based policies in the literature is MaxWeight [3, 5], which is also the earliest policy proven to be throughput-optimal in the MRJ setting [6]. Despite its throughput-optimality and low empirical mean response time, MaxWeight repeatedly solves an online optimization problem and chooses service options whenever the system state changes, leading to frequent preemption and computational inefficiency. The first nonpreemptive policy proven throughput-optimal in the MRJ setting, Randomized Multi-Resource Scheduling (RMS, also known as Randomized-Timer) [8], addresses these issues but often results in a higher empirical mean response time than MaxWeight. A family of policies called Markovian Service Rate (MSR) was recently introduced in [15]. MSR can accommodate both preemptive and nonpreemptive settings and is throughput-optimal with bounded mean response time. Chen et al. [15] showed that nonpreemptive MSR (nMSR) achieves better empirical mean response times than RMS, especially under high load. However, nMSR has an issue: it ignores the queue-length state when switching service options, resulting in still subpar mean response time. To reduce the mean response time of nMSR in the MSJ setting, Chen et al. [13] proposed Most Servers First with Quickswap (MSFQ). MSFQ is throughput-optimal in the one-or-all case and empirically achieves better mean response time and weighted mean response time than nMSR in the general integer-valued resource requirement case. Many class-based policies achieve throughput-optimality, in contrast to index-based policies.

Class-based policies are not immediately applicable to the continuous MSJ or MRJ settings, because job requirements are continuously distributed, and the system almost never contains multiple jobs with exactly identical requirements at any given time. We will build on existing class-based policies (MaxWeight and nMSR) by using a discretization scheme, to achieve throughput-optimality in the continuous setting. For a boost to empirical performance, we also augment our discretization-based policies with an index-based Backfilling approach.

3 Model

In this section, we define single- and multi-resource queueing models with continuously distributed resources (the continuous MSJ and MRJ settings). We first present the model and notation (Sections 3.1 and 3.2), then introduce our discretization-based policies in Section 3.3 and their counterparts with Backfilling in Section 3.4.

3.1 Basic Notation

We let $\mathbb{N}$ be the set of all nonnegative integers and $[K] := \{0, 1, \dots, K\}$ be the set of all nonnegative integers not greater than K . In an Euclidean space $\mathbb{R}^d$, we let an arrowed letter denote a deterministic vector (e.g., $\vec{a}$). We let $\vec{0} = (0, \dots, 0)$ and $\vec{1} = (1, \dots, 1)$. For two vectors $\vec{a}, \vec{b} \in \mathbb{R}^d$, we say $\vec{a} \preceq \vec{b}$ if $a_i \leq b_i$ for all $1 \leq i \leq d$, and let $\vec{b} - \vec{a} = (b_1 - a_1, \dots, b_d - a_d)$ denote the difference of two vectors. Given a set $C \subseteq \mathbb{R}^d$, let $\text{ConvH}(C)$ denote the convex hull of C and $\text{int}(C)$ denote the interior of C . A function f with domain $D \subseteq \mathbb{R}^d$ is said to have central symmetry (or point reflection symmetry) with respect to a point $\vec{a} \in \mathbb{R}^d$ if for any $v \in D$, $f(v) = f(2\vec{a} - v)$.

For a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, we let $\text{supp}(f)$ denote its support. For a probability distribution π , we let $\mathcal{L}(\pi|A)$ denote the conditional distribution (law) on some measurable set A .

For an $\vec{K} = (K_1, \dots, K_d) \in \mathbb{N}^d$, we let $\Gamma_{\vec{K}} := \{\vec{i} \in \mathbb{N}^d : \vec{1} \preceq \vec{i} \preceq \vec{K}\}$ and let $|\Gamma_{\vec{K}}| := |\Gamma_{\vec{K}}| = \prod_{l=1}^d K_l$.

For a vector $Z \in \mathbb{R}^{\Gamma_{\vec{K}}}$ with multi-dimensional indices, we let $Z^{(\vec{i})}$ denote the $\vec{i}$ -th element of Z , for $\vec{i} \in \Gamma_{\vec{K}}$.

In our paper, we always let $M \in \mathbb{N}^{\Gamma_{\vec{K}}}$ denote a service option, where $M^{(\vec{i})}$ is the number of type- $\vec{i}$ jobs (See Section 3.3.1) served by M . However, when $\vec{K}$ is (entry-wise) large and the vector is sparse, which happens frequently in our setting, it is inconvenient to explicitly write out M . Hence, we provide two equivalent notations for M .

Definition 1 (Alternative Notation I for $M \in \mathbb{N}^{\Gamma_{\vec{K}}}$) *Any $M \in \mathbb{N}^{\Gamma_{\vec{K}}}$ can be equivalently expressed as an unordered list of non-zero elements with repetitions. To be specific, if $\{\vec{i}_1, \vec{i}_2, \dots, \vec{i}_n\} := \{\vec{i} \in \Gamma_{\vec{K}} : M^{(\vec{i})} \neq 0\}$ are the non-zero entries of M , then we can equivalently write M such that*

$$M = \left\{ \underbrace{\vec{i}_1, \dots, \vec{i}_1}_{M^{(\vec{i}_1)} \text{'s}}, \underbrace{\vec{i}_2, \dots, \vec{i}_2}_{M^{(\vec{i}_2)} \text{'s}}, \dots, \underbrace{\vec{i}_n, \dots, \vec{i}_n}_{M^{(\vec{i}_n)} \text{'s}} \right\}. \quad (1)$$

Note that the order of elements does not matter for the uniqueness of M .

Definition 2 (Alternative Notation II for $M \in \mathbb{N}^{\Gamma_{\vec{K}}}$) *Any $M \in \mathbb{N}^{\Gamma_{\vec{K}}}$ can be equivalently expressed as a list of pairs of unique elements and their multiplicities. To be specific, if $\{\vec{i}_1, \vec{i}_2, \dots, \vec{i}_n\} := \{\vec{i} \in \Gamma_{\vec{K}} : M^{(\vec{i})} \neq 0\}$ are the non-zero entries of M , then we can equivalently write M such that*

$$M = \left\{ (\vec{i}_1, M^{(\vec{i}_1)}), (\vec{i}_1, M^{(\vec{i}_1)}), \dots, (\vec{i}_n, M^{(\vec{i}_n)}), (\vec{i}_n, M^{(\vec{i}_n)}) \right\}. \quad (2)$$

Note that the order of pairs does not matter for the uniqueness of M .

We use $M = (1, 0, 2, 0, 1, 0, 0, 0) \in \mathbb{N}^8$ as an example. Based on the alternation notation I given by Definition 1, an equivalent expression of M is $\{1, 3, 3, 5\}$. Based on the alternation notation II given by Definition 2, an equivalent expression of M is $\{(1, 1), (3, 2), (5, 1)\}$.

For the list of all notation, see Appendix A.

3.2 Multiresource-Job Queueing System with Continuous Resource Requirements

We study a model where a server has d distinct resources available, and each job requires some amount of each resource. Without loss of generality, we normalize the server's available capacity for each resource to be 1. A set of jobs can be served simultaneously whenever, for each resource type, the total requirement of these jobs does not exceed the capacity 1.

Job arrivals follow a Poisson arrival process, with arrival rate λ . Resource requirements are sampled i.i.d. from the underlying distribution with random variable V , which takes values in $(0, 1]^d$. In addition to a resource requirement, each job has a duration demand, sampled i.i.d. from a distribution with random variable D . The states of the system are lists of (requirement, duration) pairs, ordered according to their arrival times. A scheduling policy selects a set of jobs to be served simultaneously, and is assumed to be a Markovian policy, which depends only on the current state. We consider both the preempt-resume and nonpreemptive settings. When we refer to a queueing system as stable, formally we mean that the empty state is positive recurrent.

In this paper, we do not assume the resource is pre-discretized. This is in contrast to prior work, which typically assumes the job resource requirements distribution is discrete with finite support. Our assumption is more useful for real-world cloud and cluster scheduling. Specifically, resource requirement distribution V is characterized by the probability density function (p.d.f.) $f_V : (0, 1]^d \rightarrow \mathbb{R}$. For simplicity, we focus on the case where the duration D follows the exponential distribution $Exp(1)$, and the scheduling policy does not observe the remaining duration demand. However, we believe our policy should perform well under the general duration settings.

Under these assumptions, the state space simplifies to the set of current resource requirements, denoted by

$$\mathcal{S} := \left\{ x = (\vec{v}_n)_{1 \leq n \leq N} \subseteq (0, 1]^d : N \in \mathbb{N} \right\},$$

where the smaller index n means an earlier arrival time. A Markovian policy can then be defined as follows.

Definition 3 A Markovian policy π maps each state x to a subset $x' \subset x$, such that

$$\sum_{\ell=1}^{|x'|} x'_\ell \preceq \vec{1} \quad (3)$$

The subset x' refers to the set of jobs the server serves simultaneously. We define $\mathcal{S}$ to be the set of all schedulable service options x' .

Thereafter, all policies we will discuss are Markovian policies. To study the performance of a policy, we apply the drift method by specifying a Lyapunov function (test function) and analyzing its drift. Specifically, a Lyapunov function $L : \mathcal{S} \rightarrow \mathbb{R}_0^+$ maps each state of to a nonnegative number. The drift of a Lyapunov function L under some policy π , $\Delta_\pi \circ L(x)$, is then defined as the instantaneous change rate of the L at the state x ,

$$\Delta_\pi \circ L(x) = \lim_{t \downarrow 0} \frac{1}{t} \mathbb{E} [L(x(t)) - L(x(0)) \mid x(0) = x]. \quad (4)$$

3.3 New Discretization-Based Scheduling Policies

In Section 3.3.1, we first introduce our method of discretizing the resource requirements, and provide corresponding notation. We then introduce our discretization-based scheduling policies in the $\vec{K}$ -discretization system: $\vec{K}$ -discretized MaxWeight (Section 3.3.2), the $\vec{K}$ -discretized efficient MaxWeight family (Section 3.3.3), $\vec{K}$ -discretized nonpreemptive Markovian Service Rate (Section 3.3.4), and $\vec{K}$ -discretized efficient nonpreemptive Markovian Service Rate (Section 3.3.5).

3.3.1 The $\vec{K}$ -Discretization System

We begin with the single-resource queueing model, where $V \in (0, 1]$. After fixing the discretization parameter $K \in \mathbb{N}$, we equally partition the resource range $(0, 1]$ such that

$$(0, 1] = \bigcup_{k=1}^K I_k, \quad \text{where } I_k := \left(\frac{k-1}{K}, \frac{k}{K} \right], \quad \text{for } k = 1, \dots, K, \quad (5)$$

where I_k is the interval in the continuous requirement space corresponding to a single discrete requirement. We also call each interval I_k a “bucket”. Given a job with resource requirement v in the set I_k , where $k = \lceil Kv \rceil$, we say the job is a type- k job. At time t , let $q(t) = (q_1(t), \dots, q_K(t)) \in \mathbb{N}^K$ such that $q_k(t)$ denotes the number of type- k jobs in the system. We call a policy “ K -discretized” if the policy uses only the information of the job types. Therefore, under a K -discretized policy, the process $\{q(t)\}_{t \geq 0}$ forms a continuous-time Markov chain (CTMC).

Let

$$M(t) = (M^{(1)}(t), \dots, M^{(K)}(t)) \in [K]^K$$

denote the service option the server chooses at time t , where $M^{(k)}(t)$ denotes the number of type- k jobs in service, for each $k = 1, \dots, K$. Restricted by the capacity of

the server, our service option should satisfy that $M(t) \in C_K$, the set of all admissible service options, which is defined as

$$C_K := \{M \in [K]^K : \langle M, (1, \dots, K) \rangle = \sum_{k=1}^K kM^{(k)} \leq K\}. \quad (6)$$

The d -resource setting follows similarly. By fixing some $\vec{K} := (K_1, \dots, K_d) \in \mathbb{N}^d$, recall that we let $\Gamma_{\vec{K}} := \{i \in \mathbb{N}^d : \vec{1} \preceq \vec{i} \preceq \vec{K}\}$ denote the set of all job types. We partition the set $(0, 1]^d$ such that

$$(0, 1]^d = \bigcup_{\vec{i} \in \Gamma_{\vec{K}}} I_{\vec{i}} \quad \text{where } I_{\vec{i}} := \left(\frac{i_1 - 1}{K_1}, \frac{i_1}{K_1} \right] \times \dots \times \left(\frac{i_d - 1}{K_d}, \frac{i_d}{K_d} \right] \quad \text{for all } \vec{i} \in \Gamma_{\vec{K}}. \quad (7)$$

Each $I_{\vec{i}}$ is a d -dimensional box in the continuous requirement space, corresponding to a single discrete requirement. Given a job with resource requirements $\vec{v} = (v_1, \dots, v_d)$ in the set $I_{\vec{i}}$, where $\vec{i} = (\lceil K_1 v_1 \rceil, \dots, \lceil K_d v_d \rceil)$, we say the job is a type- $\vec{i}$ job. Similarly to the single-resource setting, we let $q(t) = (q_{\vec{i}}(t))_{\vec{i} \in \Gamma_{\vec{K}}}$, where $q_{\vec{i}}(t)$ denotes the number of type- $\vec{i}$ jobs in the system at time t . We similarly define a $\vec{K}$ -discretized policy to only use the information of the job types. Specifically, we let $M(t) \in \mathbb{N}^{\Gamma_{\vec{K}}}$ denote the service option at time t . Note that $M(t) \in C_{\vec{K}}$, the set of all schedulable options in $\vec{K}$ -discretized system, such that

$$C_{\vec{K}} := \{M \in [K_1] \times \dots \times [K_d] : \sum_{\vec{i} \in \Gamma_{\vec{K}}} i_l M^{(\vec{i})} \leq K_l \text{ for all } 1 \leq l \leq d\}, \quad (8)$$

where $C_{\vec{K}}$ is the set of all admissible service options under $\vec{K}$ -discretization, and $M^{(\vec{i})}(t)$ denotes the number of type- $\vec{i}$ jobs in that are served at time t .

We now provide some definitions on arrival and service measures, before we move to introducing our novel discretization-based policies. We recall that the system has the arrival rate λ and the job resource requirement distribution $V \in (0, 1]^d$ with density $f_V : (0, 1]^d \rightarrow \mathbb{R}$.

Definition 4 ($(\vec{K}$ -Discretized) Arrival Measures) *Given the arrival rate λ and the resource requirement density f_V , the **arrival measure** η^A on the space $(0, 1]^d$ is given by*

$$\eta^A(dx) := \lambda f_V(dx). \quad (9)$$

*Moreover, for any fixed $\vec{K} \in \mathbb{N}^d$, the **$\vec{K}$ -discretized arrival measure** ($\vec{K}$ -arrival measure) $\eta_{\vec{K}}^A$ is defined over the set $\Gamma_{\vec{K}}$ of all job types, by integrating the density over the box $I_{\vec{i}}$, such that*

$$\eta_{\vec{K}}^A(\vec{i}) := \eta^A(I_{\vec{i}}) = \int_{I_{\vec{i}}} \lambda f_V(x) dx, \quad \forall \vec{i} \in \Gamma_{\vec{K}}. \quad (10)$$

*The **$\vec{K}$ -arrival rate vector** $\Lambda_{\vec{K}} \in \mathbb{R}^{\Gamma_{\vec{K}}}$ with indices in $\Gamma_{\vec{K}}$, which characterizes $\eta_{\vec{K}}^A$, is then defined such that $\Lambda_{\vec{K}}(\vec{i}) = \eta_{\vec{K}}^A(\vec{i})$.*

Definition 5 (Service Option Distribution and Discretized Service Measure) *For any fixed $\vec{K}$ -discretization, we obtain a set of all admissible service options $C_{\vec{K}}$ (8). We let $\beta_{\vec{K}}$ denote a generic service option distribution over $C_{\vec{K}}$. The measure induced by $\beta_{\vec{K}}$ is defined over the set $\Gamma_{\vec{K}}$, the mass of which is given by*

$$\eta_{\vec{K}}^S(\vec{i}) = \mathbb{E}^{\beta_{\vec{K}}} [M^{(\vec{i})}] = \sum_{M \in C_{\vec{K}}} \beta_{\vec{K}}(M) M^{(\vec{i})}, \quad \forall \vec{i} \in \Gamma_{\vec{K}}. \quad (11)$$

We call $\eta_{\vec{K}}^S$ the **discretized service measure** induced by $\beta_{\vec{K}}$.

3.3.2 The $\vec{K}$ -discretized MaxWeight Scheduling Policy

In the single-resource setting, with a chosen discretization parameter $K \in \mathbb{N}$, the K -**discretized MaxWeight scheduling policy** (K -MW) at time t chooses the service option

$$M^{MW-C_K}(t) := \arg \max_{M \in C_K} \langle M, q(t) \rangle. \quad (12)$$

Similarly in the d -resource setting, with some preset $\vec{K} \in \mathbb{N}^d$, we define the $\vec{K}$ -discretized MaxWeight scheduling policy ($\vec{K}$ -MW), which at time t chooses the service option

$$M^{MW-C_{\vec{K}}}(t) := \arg \max_{M \in C_{\vec{K}}} \langle M, q(t) \rangle = \arg \max_{M \in C_{\vec{K}}} \sum_{\vec{i} \in \Gamma_{\vec{K}}} q_{\vec{i}}(t) M^{(\vec{i})}. \quad (13)$$

We need to carefully select $\vec{K}$ by considering the tradeoff between achieving a better mean response time and obtaining a larger stability region. A large $\vec{K}$ makes our policy close to naive MaxWeight, which is equivalent to Least-Server-First until very large queue length, and leads to a worse mean response time. However, a small $\vec{K}$ causes a coarse discretization that treats every job like the requirement is maximum for the type. The waste of capacity in rounding up results in a poor stability region.

One of our main focuses is to show that the $\vec{K}$ -MW family of policies is throughput-optimal, which means if there is a policy that can stabilize the system, then there exists some $\vec{K}$ such that $\vec{K}$ -MW can also stabilize the system. Lemma 1 will provide a sufficient stability condition for $\vec{K}$ -MW. Theorem 1 will prove the throughput-optimality of this policy given that the p.d.f. of the resource requirement V is of finite Lipschitz constant.

3.3.3 $\vec{K}$ -discretized Efficient MaxWeight

One drawback of $\vec{K}$ -MW is its large candidate set, resulting in poor empirical efficiency when $\vec{K}$ is large. The cardinality of the set $C_{\vec{K}}$ of all admissible service options grows exponentially with the square root of $\max\{K_i\}$. For example, in the single-resource setting, the number of full-utilization service options is $\tilde{\theta}(\exp(\pi\sqrt{\frac{2K}{3}}))$ [32]. Therefore, when we choose a large discretization parameter $\vec{K}$, a complicated optimization problem must be solved every time the state changes.

Faced with this problem, we propose a family of policies called **$\vec{K}$ -discretized Efficient MaxWeight** ($\vec{K}$ -EMW). Each policy in this family specifies an efficient

set of service options $E_{\vec{K}} \subset C_{\vec{K}}$. The policy only searches over candidates in $E_{\vec{K}}$ and selects the following service option:

$$M^{MW-E_{\vec{K}}}(t) := \arg \max_{M \in E_{\vec{K}}} \langle M, q(t) \rangle. \quad (14)$$

We will choose the following four efficient sets that still guarantee throughput-optimality: the 2-Job efficient set E_K^{2J} , the 2-Bucket efficient set E_K^{2B} , the Extreme-vertices efficient set E_K^X , and the Pairwise Extreme-vertices (XP) efficient set E_K^{XP} .

The 2-Job efficient set E_K^{2J} : We focus on this efficient set in settings with symmetric resource distribution V . In the single-resource setting, E_K^{2J} consists of all candidates that serve one type- j job and one type- $(K-j)$ job together, and the candidate that serves one type- K job. More specifically, if K is odd, we define $E_K^{2J} = \{M_K, M_1, \dots, M_{\frac{K-1}{2}}\}$, such that $M_K = \{K\}$, and $M_j = \{j, K-j\}$ for each $j = 1, \dots, \frac{K-1}{2}$. For even K , we add the option that serves two type- $\frac{K}{2}$ jobs.

In the d -resource setting and given $\vec{K}$ -discretization, we define $E_{\vec{K}}^{2J}$ for a $\vec{K}$ discretization as follows. Recall that $\vec{K} = (K_1, \dots, K_d) \in \mathbb{N}^d$. We consider the candidates that serve one type- $\vec{j}$ job and one type- $(\vec{K} - \vec{j})$ job together, and the candidates that serve only one job that has at least one resource requirement reaching the capacity limit (“boundary” job). To be specific, assume $K_1, \dots, K_d$ to be odd. For any $\vec{j} \in \mathcal{B}^{\vec{K}}$, the boundary set, such that

$$\mathcal{B}^{\vec{K}} = \{\vec{j} : j_\ell = K_\ell \text{ for some } \ell = 1, \dots, d\}, \quad (15)$$

we set $M_{\vec{j}} = \{\vec{j}\}$. Then for each $(\frac{K_1+1}{2}, 1, \dots, 1) \preceq \vec{j} \preceq (K_1-1, \dots, K_d-1)$, we let $M_{\vec{j}} = \{\vec{j}, \vec{K} - \vec{j}\}$. We handle even $\vec{K}$ as in the $d = 1$ case.

The MaxWeight policy that uses $E_{\vec{K}}^{2J}$ as the candidate set is then called **2-Job $\vec{K}$ -Efficient MaxWeight** (2J $\vec{K}$ -EMW).

The 2-Bucket efficient set E_K^{2B} : We focus on E_K^{2B} in $d = 1$ single-resource setting, where the requirement density f_V is decreasing. We let $K = 2^L$ for some integer L . E_K^{2B} contains the following candidates, expressed by the alternative notation II:

- For $k = 2^\ell \in [K]$ that is a power-of-two integer (including $k = 1$), we include the service option that serves $2^{L-\ell}$ type- k jobs simultaneously. Specifically, $M_{2^\ell} = \{(2^\ell, 2^{L-\ell})\}$.
- For $k \in [K]$ that is not a power-of-two integer, let $\ell := \ell(k) = \lceil \log_2 j \rceil$. We include the service option that serves $2^{L-\ell}$ type- k jobs and $2^{L-\ell}$ type- $(2^\ell - k)$ jobs simultaneously. Specifically, $M_k = \{(k, 2^{L-\ell}), (2^\ell - k, 2^{L-\ell})\}$.

We call the MaxWeight policy with efficient set E_K^{2B} **2-Bucket K-Efficient Discretized MaxWeight** (2B K-EMW).

The Extreme-vertices efficient set E_K^X : E_K^X is applicable under any requirement distribution V . For any $\vec{K}$ fixed, we let $E_{\vec{K}}^X$ denote the set of all extreme vertices in $C_{\vec{K}}$ (the extreme points in $\text{ConvH}(C_{\vec{K}})$). The corresponding policy that chooses

$M^X = \arg \max_{M \in E_K^X} \langle M, q(t) \rangle$ is called **Extreme-vertices $\vec{K}$ -Efficient Discretized MaxWeight (X $\vec{K}$ -EMW)**.

The *Pairwise Extreme-vertices (XP) efficient set* E_K^{XP} : E_K^{XP} is applicable under the MSJ setting when the discretization parameter K is even. It is designed as a tractable surrogate for the X efficient set E_K^X , because the exact construction of E_K^X is computationally difficult. Specifically, E_K^{XP} is a superset of E_K^X and can be constructed by a direct set-subtraction procedure.

Formally speaking, for any $N \leq K$, we define the set $\tilde{C}_N$, which represents the set of service options with total resource usage N/K :

$$\tilde{C}_N := \{M \in \mathbb{N}^K : \sum_{k=1}^K kM^{(k)} = N\}.$$

The XP efficient set is obtained from $\tilde{C}_K$ by excluding every service option that can be decomposed into two distinct service options, each with total resource usage $1/2$:

$$E_K^{XP} := \{M \in \tilde{C}_K : M \neq M_1 + M_2, \text{ where } M_1 \neq M_2 \text{ and } M_1, M_2 \in \tilde{C}_{\frac{K}{2}}\}. \quad (16)$$

We call the MaxWeight policy using E_K^{XP} **Pairwise Extreme-vertices K -Efficient Discretized MaxWeight (XP K -EMW)**.

3.3.4 $\vec{K}$ -Discretized Nonpreemptive Markovian Service Rate

In addition to the above preemptive policies, we provide nonpreemptive policies to schedule jobs with continuous resource requirements. This family of policies is built from the nonpreemptive Markovian Service Rate (nMSR) policy [15]. MSR uses an irreducible, finite-state continuous-time Markov chain $\{M(t)\}_{t \geq 0}$ as a modulating process. The state space of $\{M(t)\}_{t \geq 0}$ is a set of service options. As proven in [15], the family of the nMSR policies is throughput-optimal. In addition, the only computational expense is the precomputation to find the modulating process – executing the process is more efficient.

In our setting, where the resource requirement distribution is continuous, we again use $\vec{K}$ -discretization as introduced in Section 3.3.1, and consider the set of all schedulable discretized service options $C_{\vec{K}}$. By conducting an offline pre-computation, we choose at most $|\Gamma_{\vec{K}}|$ service options in $C_{\vec{K}}$ to form the state space of the modulating process $\{M(t)\}_{t \geq 0}$. Then nMSR can be generalized to the continuous setting, called **$\vec{K}$ -discretized nMSR ($\vec{K}$ -nMSR)**. We will show in Theorem 2 that $\vec{K}$ -nMSR is throughput-optimal, given that the p.d.f. of the resource requirement is Lipschitz continuous.

3.3.5 $\vec{K}$ -Discretized Efficient Nonpreemptive Markovian Service Rate

To reduce the complexity of the offline computation in selecting the modulating process, the efficient set setting can be extended to the nonpreemptive case. Specifically,

after $\vec{K}$ -discretization, nMSR chooses service options from an efficient set. We recall Section 3.3.3 and define efficient nMSR policies corresponding to the 2-Job (2J $\vec{K}$ -EnMSR), 2-Bucket (2B $\vec{K}$ -EnMSR), Extreme-vertices (X $\vec{K}$ -EnMSR), and Pairwise Extreme-vertices (XP $\vec{K}$ -EnMSR) efficient sets.

3.4 Discretization-Based Policies with Backfilling

To fully utilize the available capacity, Backfilling can be applied for each preemptive discretization-based policy. Backfilling works the same way as the filling rule of First-Fit. Once the underlying policy selects a candidate service option, the remaining unused resources may still allow some jobs in the system to be packed. The server then scans the queue in arrival order and packs jobs by the following greedy rule. The server packs a scanned job and serves it together with the jobs already selected, whenever the resource requirements of the scanned job are at most the remaining resources of the server. For brevity, we refer to $\vec{K}$ -discretized MaxWeight with Backfilling as $\vec{K}$ -MW-B. We note that $\vec{K}$ -MW-B or $\vec{K}$ -EMW-B are not purely discretization-based policies.

Lemma 3 states that the same type of proof of stability for $\vec{K}$ -MW (or $\vec{K}$ -EMW) also guarantees the stability of $\vec{K}$ -MW-B (or $\vec{K}$ -EMW-B). In addition, adopting Backfilling improves performance without significantly worsening computation efficiency, which is verified in Section 6.

4 Main Results

Our main goal is to find the first throughput-optimal family of policies in the continuous MRJ setting. We begin by stating our core result: In both single- and multi-resource settings, the family of $\vec{K}$ -discretized MaxWeight ($\vec{K}$ -MW) policies is throughput-optimal for all resource distributions V with Lipschitz continuous p.d.f.s f_V .

Theorem 1 (Throughput-optimality of $\vec{K}$ -MW) *In the continuous MRJ setting with exponential durations, for any resource requirement distribution V with Lipschitz continuous probability density function f_V over $(0, 1]^d$, the family of $\vec{K}$ -MW policies is throughput-optimal. That is, if the continuous MRJ system is stable under some policy π , then there exists some $\vec{K}$ such that $\vec{K}$ -MW stabilizes the system.*

The proof is deferred to Section 5.2.

Proof Sketch: We show that under the discretized $\vec{K}$ -MW policy, we can switch our focus to the discretized CTMC q_t introduced in Section 3.3.1. We build upon the standard stability argument for class-based policies: It suffices to exhibit a distribution $\beta_{\vec{K}}$ over service options that fulfills a discrete measure dominance condition. The condition states that the total service rate for each class under $\beta_{\vec{K}}$ exceeds its arrival rate (Lemma 1).

The challenge arises because the conversion to the discrete system involves rounding up the resource requirements, increasing the system load. As a result, the system

may fail to be stable. To overcome this, we must both choose $\vec{K}$ carefully based on the Lipschitz constant of f_V , as well as carefully construct $\beta_{\vec{K}}$.

To construct $\beta_{\vec{K}}$ satisfying discrete measure dominance, we use the fact that there exists a stable continuous policy π , which induces a continuous measure dominance for the undiscretized system. However, the naive construction of $\beta_{\vec{K}}$ fails. Simply rounding up each continuous service option leads to inadmissible service options. We instead round *down* each continuous service option to generate admissible service options, and invoke Lipschitz continuity of f_V , to argue that the density of the rounded-up and rounded-down buckets are similar. Our approach requires special care for certain job types, including

- Job types that use the entirety of some resource,
- Job types on the boundary of the support of V , and
- Low-density job types.

By carefully handling these job types, we demonstrate that we can always choose a $\vec{K}$ and a $\beta_{\vec{K}}$ such that the discrete measure dominance is satisfied, and conclude the theorem. $\square$

When jobs are not preemptible, a nonpreemptive family of policies is needed. In Theorem 2, we show that the family of $\vec{K}$ -nMSR policies is the first nonpreemptive throughput-optimal family in the continuous MRJ setting, under the same mild assumption of Theorem 1.

Theorem 2 (Throughput-optimality of $\vec{K}$ -nMSR) *In the continuous MRJ setting, the family of $\vec{K}$ -nMSR policies is throughput-optimal for all V with Lipschitz continuous density f_V .*

The proof is deferred to Section 5.3. The proof follows similar steps to Theorem 1, because the stability condition of $\vec{K}$ -nMSR (Lemma 5) is equivalent to that of $\vec{K}$ -MW, as we described in Lemma 2.

The candidate set $C_{\vec{K}}$ of all schedulable $\vec{K}$ -discretized service options is extremely large for large $\vec{K}$. To execute $\vec{K}$ -MW, the server has to solve a complicated optimization problem every time the state changes. Similarly, to execute $\vec{K}$ -nMSR, the server requires to perform an expensive offline pre-computation to select the modulating process.

To avoid the computation expense of searching over all of $C_{\vec{K}}$, we provide more efficient families of policies by considering smaller subsets of candidate service options. These subsets of $C_{\vec{K}}$ are called efficient sets. Recall first the extreme-vertices set $E_{\vec{K}}^X \subset C_{\vec{K}}$ from Section 3.3.3, which is the set of all extreme points of $\text{ConvH}(C_{\vec{K}})$. The following theorem states that the corresponding discretized MaxWeight and nMSR policies are throughput-optimal under the same assumption as Theorem 1.

Theorem 3 (Throughput-optimality of X $\vec{K}$ -EMW and X $\vec{K}$ -EnMSR) *In the continuous MRJ setting, given the same conditions as in Theorem 1, X $\vec{K}$ -EMW and X $\vec{K}$ -EnMSR are both throughput-optimal. Moreover, for any $\vec{K} \in \mathbb{N}^d$ fixed, $\vec{K}$ -MW, X $\vec{K}$ -EMW, $\vec{K}$ -nMSR, and X $\vec{K}$ -EnMSR all are stable for the same set of arrival rates.*

The proof of Theorem 3 is deferred to Section 5.4. The efficiency of $X \vec{K}$ -EMW and $X \vec{K}$ -EnMSR lies in the fact that the ratio $|E_K^X|/|C_K|$ vanishes when $\vec{K}$ grows. For example, in the single-resource setting, when $K = 60$, both X efficient policies consider 0.53% as many service options as 60-MW, and when $K = 100$, the ratio drops to 0.031% (Shlyk [33]). However, while E_K^X significantly reduces the number of service option candidates, its cardinality still goes superpolynomially in $\max(\vec{K})$, so there is still the room for efficiency.

In the search for yet more efficient policies, we turn to a simpler resource distribution in the MSJ (single-dimensional) setting. We consider resource distributions V such that f_V is weakly decreasing. This allows us to achieve throughput-optimality with an efficient set, whose size is K , in contrast to the super-polynomial candidate set sizes of C_k and E_K^X . Recall the 2-bucket efficient sets from Section 3.3.3, with corresponding policies 2B K -EMW and 2B K -EnMSR. We show both are throughput-optimal in this decreasing density setting.

Theorem 4 (2B K -EMW and 2B K -EnMSR) *In the continuous MSJ setting with the weakly decreasing density function of V , the 2B K -EMW and 2B K -EnMSR families are throughput-optimal. In particular, for any $\lambda < \frac{1}{\mathbb{E}V}$, there exists some K such that the queuing system is stable under 2B K -EMW and under 2B K -EnMSR. In addition, it is enough to let $K = 2^L$, where $L = \lfloor -\log_2(\frac{1}{\lambda} - \mathbb{E}V) \rfloor + 1$.*

The proof of Theorem is deferred to Section 5.6.

Proof Sketch: We first derive the upper bound of the stability region in Lemma 7, by showing no policy is stable when $\lambda \geq \frac{1}{\mathbb{E}V}$. Next, we use Lemma 1 to prove that when $\lambda < \frac{1}{\mathbb{E}V}$, there exists some K such that both 2B K -EMW and 2B K -nMSR are stable. The key idea in our construction of $\beta_{\vec{K}}$ over E_K^{2B} to achieve discrete measure dominance is to greedily assign mass to each service option in a clever order. We assign mass starting with service options serving jobs with higher resource requirements down to lower requirements. The weakly decreasing f_V allows us to construct a $\beta_{\vec{K}}$ distribution over service options, guaranteeing discrete measure dominance while only using service options that fully utilize all resources. $\square$

Theorem 4 does not need to assume any continuity condition for f_V Lipschitz or otherwise. Our choice of K is fully explicit, depending only on the arrival rate λ and the expectation $\mathbb{E}V$.

We then focus on another setting, where the resource requirement distribution V is centrally symmetric and is of arbitrary dimension. We have the 2-Job efficient set E_K^{2J} , introduced in Section 3.3.3. E_K^{2J} contains $\prod \vec{K}/2$ service options, half as many as the number of job types. Recall from Sections 3.3.3 and 3.3.5, we call the corresponding efficient MaxWeight policy and nMSR policy 2J $\vec{K}$ -EMW and 2J $\vec{K}$ -EnMSR, respectively. Theorem 5 shows the throughput-optimality result for 2J $\vec{K}$ -EMW and 2J $\vec{K}$ -EnMSR, for resource requirement V such that its p.d.f. f_V is Lipschitz continuous and centrally symmetric with respect to $\frac{1}{2}\vec{1}$.

Theorem 5 (2J $\vec{K}$ -EMW and 2J $\vec{K}$ -EnMSR) *In the continuous MRJ setting, assume that the p.d.f. f_V is Lipschitz continuous and centrally symmetric with respect to $\frac{1}{2}\vec{1}$. In this case, both the 2J $\vec{K}$ -EMW and 2J $\vec{K}$ -EnMSR families are throughput-optimal. To be specific, for any arrival rate $\lambda < 2$, there exists some $\vec{K}$ such that both 2J $\vec{K}$ -EMW and 2J $\vec{K}$ -EnMSR stabilize the system.*

Moreover, if $V \sim \text{Unif}((0, 1]^d)$, any $\vec{K} \succeq K\vec{1}$ stabilizes the system, where K is the smallest odd integer such that $K \geq \lfloor \frac{2\lambda d}{2-\lambda} \rfloor + 1$. Additionally, in the single-resource setting, it suffices to choose an odd K such that $K \geq \lfloor \frac{\lambda}{2-\lambda} \rfloor + 1$.

The proof of Theorem 5 is deferred to Section 5.7.

Proof Sketch: On the one hand, we use Lemma 7 to show that no policy is stable when $\lambda \geq 2$. On the other hand, for any $\lambda < 2$, we construct discrete measure dominance and apply Lemma 1. The key idea is to assign mass to the option serving the type- $\vec{i}$ jobs and the type- $(\vec{K} - \vec{i})$ equal to the larger arrival rate of the two job types. The two arrival rates differ negligibly due to the symmetry Lipschitz condition of f_V . We also separately handle the boundary set, involving jobs that fully utilize a resource. By choosing an appropriately large $\vec{K}$, which depends both on the Lipschitz constant and the size of the boundary set, we construct a distribution $\beta_{\vec{K}}$ satisfying the discrete measure dominance condition, similar to the proof of Theorem 1. $\square$

In the single-dimensional (MSJ) setting, the efficient set has cardinality $|E_K^{2J}| = \frac{K+1}{2}$, which grows linearly in K , compared with $|C_K|$'s exponential growth in K . In the multi-dimensional (MRJ) setting, when we let $\vec{K} = K\vec{1}$, the cardinality of the 2-Job K -efficient set $|E_{\vec{K}}^{2J}|$ grows exponentially in d and polynomially in K , whereas $|C_{\vec{K}}|$ grows exponentially both in K and d .

We note in Lemma 3, that our stability condition for a MaxWeight-type policy also suffices to demonstrate stability when Backfilling is incorporated, as described in Section 3.4. Consequently, under the same conditions as Theorems 1 and 3 to 5, the corresponding families of $\vec{K}$ -MW-B and $\vec{K}$ -EMW-B are also throughput optimal. In Section 6, we further examine the empirical performance. In addition, we empirically demonstrate that our policies achieve the state-of-the-art performance, both on parametric distribution and datacenter traces.

5 Proofs

In this section, we provide the full proofs of our main results in Section 4.

In Section 5.1, we start by proving a general sufficient condition for stability (Lemma 1) for $\vec{K}$ -discretized MaxWeight ($\vec{K}$ -MW) and Efficient $\vec{K}$ -MW ($\vec{K}$ -EMW), with given $\vec{K}$ and efficient set $E_{\vec{K}}$. We also show in Lemma 3 that this condition suffices for stability for $\vec{K}$ -MW with Backfilling ($\vec{K}$ -MW-B) and $\vec{K}$ -EMW with Backfilling ($\vec{K}$ -EMW-B).

In Section 5.2, we prove our first main result, Theorem 1, demonstrating throughput-optimality for the $\vec{K}$ -MW family. To do so, we state and prove Lemma 4, which guarantees the existence of a discretization parameter $\vec{K}$ based on the arrival measure satisfying the stability condition in Lemma 1. We then use Lemmas 4 and 6 to show the throughput-optimality of the $\vec{K}$ -MW family (Theorem 1).

In Section 5.3, we connect the sufficient stability condition in Lemma 1 with the existing stability condition of $\vec{K}$ -nonpreemptive Markovian Service Rate ($\vec{K}$ -nMSR) [15]. We then combine the stability condition with Lemma 4 and show the throughput-optimality of the $\vec{K}$ -nMSR family (Theorem 2).

In Section 5.4, we prove Theorem 3, which states the throughput-optimality of the Extreme-vertices $\vec{K}$ -EMW (X $\vec{K}$ -EMW) family and X $\vec{K}$ -EnMSR family. We also show the throughput-optimality of the Pairwise Extreme-vertices $\vec{K}$ -EMW (XP $\vec{K}$ -EMW) family and XP $\vec{K}$ -EnMSR family, which is a consequence of our proof.

In Section 5.5, we prove Lemma 7, which gives an upper bound of the stability region for any MRJ system. In Section 5.6, we prove the 2-Bucket K -EMW (2B K -EMW) family and the 2B K -EnMSR family, when V has single dimensional decreasing density (Theorem 4). Finally, in Section 5.7, we focus on symmetrical distribution of V and show the throughput-optimality of the 2-Job $\vec{K}$ -EMW (2J $\vec{K}$ -EMW) family and the 2J $\vec{K}$ -EnMSR family (Theorem 5). The core idea is that, for any arrival rate below the upper bound given in Lemma 7, we explicitly construct some $\vec{K}$ and an appropriate distribution $\beta_{\vec{K}}$ over each efficient set, such that the stability conditions are satisfied.

5.1 Sufficient Stability Condition for $\vec{K}$ -MW and $\vec{K}$ -EMW Families

We now provide our sufficient stability condition for $\vec{K}$ -MW and $\vec{K}$ -EMW.

Lemma 1 *In the continuous MRJ setting, for a fixed $\vec{K} \in \mathbb{N}^d$, the $\vec{K}$ -MW policy is stable if there exists a service option distribution $\beta_{\vec{K}}$ such that its corresponding service measure $\eta_{\vec{K}}^S$ (see Definition 5), satisfies the following discrete measure dominance condition*

$$\eta_{\vec{K}}^S(\vec{i}) > \eta_{\vec{K}}^A(\vec{i}), \quad \forall \vec{i} \in \Gamma_{\vec{K}}. \quad (17)$$

Moreover, the $\vec{K}$ -EMW policy with efficient set $E_{\vec{K}} := \{M_j\}_{j \in \mathcal{J}}$ is stable if the condition (17) holds, and the support of $\beta_{\vec{K}}$ is a subset of $E_{\vec{K}}$.

Proof Given that the measure dominance condition (17) holds, we will use the Foster-Lyapunov Theorem [34] to prove the stability of $\{q(t)\}_{t \geq 0}$, the CTMC defining the $\vec{K}$ -discretized system in Section 3.3.1. In this context, we define the Lyapunov function $L : \mathbb{N}^{\Gamma_{\vec{K}}} \rightarrow \mathbb{R}$ such that

$$L(q) = \|q\|_2^2 = \sum_{\vec{i} \in \Gamma_{\vec{K}}} q_{\vec{i}}^2. \quad (18)$$

We let $\Delta_M \circ L(q)$ denote the drift (4) of $L(q)$ under our K -EMW policy for some efficient set, which also applies to the full K -MW. The Foster-Lyapunov Theorem states that the CTMC $\{q(t)\}_{t \geq 0}$ is positive recurrent if there exist constants $c > 0$, $b > 0$ and a finite set $C \subset \mathbb{N}^{\Gamma_{\vec{K}}}$ such that

$$\Delta_M \circ L(q) \leq -c + b \mathbf{1}\{q \in C\}, \quad \text{for all } q \in \mathbb{N}^{\Gamma_{\vec{K}}}. \quad (19)$$

We define the random vector $A(t) := (A_{\vec{i}}(t)) \in \mathbb{N}^{\Gamma_{\vec{K}}}$, where for each job type $\vec{i} \in \Gamma_{\vec{K}}$, the entry $A_{\vec{i}}(t)$ denotes the numbers of type- $\vec{i}$ jobs arriving in the time period $(0, t]$. We define

the potential completion $S(t) \in \mathbb{N}^{\Gamma_{\vec{K}}}$, and the realized completion $S(t) \in \mathbb{N}^{\Gamma_{\vec{K}}}$ similarly, such that $\tilde{S}_{\vec{i}}(t)$ and $S_{\vec{i}}(t)$ denote the number of type- $\vec{i}$ jobs our policy attempts to complete and actually completes, respectively.

We now clarify the distinction between $\tilde{S}_{\vec{i}}(t)$ and $S_{\vec{i}}(t)$. During a time period with service option M that attempts to serve $M^{(\vec{i})}$ type- $\vec{i}$ jobs, the potential completion process of job type $\vec{i}$ is a Poisson process $PP(M^{(\vec{i})})$. However, the system may have less than $M^{(\vec{i})}$ type- $\vec{i}$ jobs during this period, resulting in the realized completions $\tilde{S}_{\vec{i}}(t)$ being less than the potential completions $S_{\vec{i}}(t)$. Hence, we note that $\tilde{S}_{\vec{i}}(t) \leq S_{\vec{i}}(t)$ because some potential service may go unused.

We now upper bound the drift $\Delta_M \circ L$ of our Lyapunov function. Given any $q \in \mathbb{N}^{\Gamma_{\vec{K}}}$,

$$\begin{aligned}
\Delta_M \circ L(q) &= \lim_{t \downarrow 0} \frac{1}{t} \mathbb{E}[V(q(t)) - V(q(0)) \mid q(0) = q] \\
&= \lim_{t \downarrow 0} \frac{1}{t} \mathbb{E} \left[\left\| q(0) + A(t) - \tilde{S}(t) \right\|_2^2 - \|q(0)\|_2^2 \mid q(0) = q \right] \\
&= \lim_{t \downarrow 0} \frac{1}{t} \mathbb{E} \left[\left\| A(t) - \tilde{S}(t) \right\|_2^2 \mid q(0) = q \right] + 2 \lim_{t \downarrow 0} \frac{1}{t} \mathbb{E} \left[\langle q(0), A(t) - \tilde{S}(t) \rangle \mid q(0) = q \right] \\
&= \lim_{t \downarrow 0} \frac{1}{t} \left\{ \underbrace{\mathbb{E} \left[\left\| A(t) - \tilde{S}(t) \right\|_2^2 \mid q(0) = q \right]}_{(I)} + 2 \underbrace{\mathbb{E} \left[\langle q(0), A(t) - \tilde{S}(t) \rangle \mid q(0) = q \right]}_{(II)} \right. \\
&\quad \left. + 2 \underbrace{\mathbb{E} \left[\langle q(0), S(t) - \tilde{S}(t) \rangle \mid q(0) = q \right]}_{(III)} \right\} \tag{20}
\end{aligned}$$

We now upper bound (I), (II), and (III). For (I), by letting $K_m := \min\{K_1, \dots, K_d\}$, we have that

$$\begin{aligned}
(I) &= \mathbb{E} \left[\sum_{\vec{i} \in \Gamma_{\vec{K}}} \left(A_{\vec{i}}(t) - \tilde{S}_{\vec{i}}(t) \right)^2 \mid q(0) = q \right] \\
&= \sum_{\vec{i} \in \Gamma_{\vec{K}}} \mathbb{E} A_{\vec{i}}^2(t) + \mathbb{E} \left[\sum_{\vec{i} \in \Gamma_{\vec{K}}} \tilde{S}_{\vec{i}}^2(t) \mid q(0) = q \right] - 2 \sum_{\vec{i} \in \Gamma_{\vec{K}}} \mathbb{E} A_{\vec{i}}(t) \mathbb{E} [\tilde{S}_{\vec{i}}(t) \mid q(0) = q] \\
&\leq \mathbb{E} \left[\left(\sum_{\vec{i} \in \Gamma_{\vec{K}}} A_{\vec{i}}(t) \right)^2 \right] + \mathbb{E} \left[\left(\sum_{\vec{i} \in \Gamma_{\vec{K}}} \tilde{S}_{\vec{i}}(t) \right)^2 \mid q(0) = q \right] \tag{21}
\end{aligned}$$

$$\leq \mathbb{E} \left[\left(\sum_{\vec{i} \in \Gamma_{\vec{K}}} A_{\vec{i}}(t) \right)^2 \right] + \mathbb{E} \left[\left(\sum_{\vec{i} \in \Gamma_{\vec{K}}} S_{\vec{i}}(t) \right)^2 \mid q(0) = q \right] \tag{22}$$

$$\leq \lambda t + \lambda^2 t^2 + K_m t + K_m^2 t^2. \tag{23}$$

where

- (21) is due to $A_{\vec{i}}(t)$ and $\tilde{S}_{\vec{i}}(t)$ are nonnegative for all $\vec{i}$,
- (22) is due to $\tilde{S}(t) \preceq S(t)$, and
- (23) is due to $\sum_{\vec{i} \in \Gamma_{\vec{K}}} A_{\vec{i}}(t) \sim \text{Poisson}(\lambda t)$ and $\sum_{\vec{i} \in \Gamma_{\vec{K}}} S_{\vec{i}}(t)$ is stochastically dominated by $\text{Poisson}(K_m t)$, because the server can serve at most K_m jobs simultaneously, which in turn holds because the minimum resource m requirement is $1/K_m$.

Hence, $\lim_{t \downarrow 0} \frac{(I)}{t} \leq \lambda + K_m$. Note that this bound holds for any $\vec{K}$ -discretized policy.

For (II), we primarily focus on $\vec{K}$ -EMW policy (13) with some efficient set $E_{\vec{K}}$. $\vec{K}$ -MW can be treated as a special case where the efficient set $E_{\vec{K}} = C_{\vec{K}}$.

The corresponding service process is approximated by an entry-wise independent Poisson distribution over short time intervals, such that $S^{EMW}(t) \sim \text{Poisson}(M^{MW-E_{\vec{K}}}(0)t) + \epsilon_t$, where ϵ_t satisfies $\lim_{t \downarrow 0} \mathbb{P}(\|\frac{\epsilon_t}{t}\|_2 > \delta) = 0$ for any $\delta > 0$. In addition, the discrete measure

dominance condition (17) means that there exist some positive δ such that for all $\vec{i} \in \Gamma_{\vec{K}}$,

$$\eta_{\vec{K}}^S(\vec{i}) = \sum_{j \in \mathcal{J}} \beta_j M_j^{(\vec{i})} \geq (1 + \delta) \eta_{\vec{K}}^A = (1 + \delta) \Lambda_{\vec{K}}(\vec{i}). \quad (24)$$

Therefore, term (II) has the following upper bound:

$$\begin{aligned} (II) &= \langle q, \Lambda_{\vec{K}} t - M^{MW-E_{\vec{K}}}(0)t + o(t) \rangle \\ &= \langle q, \Lambda_{\vec{K}} t + o(t) \rangle - \langle q, t \sum_{j=1}^J \beta_j M^{MW-E_{\vec{K}}}(0) \rangle \\ &= \langle q, \Lambda_{\vec{K}} t + o(t) \rangle - t \sum_{j=1}^J \beta_j \langle q, M^{MW-E_{\vec{K}}}(0) \rangle \\ &\leq \langle q, \Lambda_{\vec{K}} t + o(t) \rangle - t \sum_{j=1}^J \beta_j \langle q, M_j \rangle \\ &\leq \langle q, \Lambda_{\vec{K}} t + o(t) \rangle - \langle q, (1 + \delta) \Lambda_{\vec{K}} t \rangle \\ &= \langle q, -\delta \Lambda_{\vec{K}} t + o(t) \rangle, \end{aligned} \quad (25)$$

where inequality (25) is due to the definition of $\vec{K}$ -EMW, and inequality (26) is due to (24). As a result,

$$\lim_{t \downarrow 0} (II)/t \leq -\delta \langle q, \Lambda_{\vec{K}} \rangle \leq \delta \lambda_{\min} \|q\|_1,$$

where $\lambda_{\min} = \min\{\mathbb{P}(V \in I_{\vec{i}}) : \mathbb{P}(V \in I_{\vec{i}}) > 0, \vec{i} \in \Gamma_{\vec{K}}\}$ is the smallest positive arrival rate for any job type $\vec{i}$. Here we disregard the buckets $I_{\vec{i}}$ with $\mathbb{P}(V \in I_{\vec{i}}) = 0$ because it is almost never that $q(\vec{i}) > 0$ occurs for such buckets.

Finally, for item (III) we again consider $\vec{K}$ -MW as a special case of $\vec{K}$ -EMW. We have (III) = $\sum_{\vec{i} \in \Gamma_{\vec{K}}} \mathbb{E}[q_{\vec{i}}(S_{\vec{i}}(t) - \tilde{S}_{\vec{i}}(t)) | q(0) = q]$. For each job type $\vec{i}$, notice that $\lim_{t \downarrow 0} \frac{\mathbb{P}(S_{\vec{i}}(t) - \tilde{S}_{\vec{i}}(t) > 0)}{t} > 0$ only if the queue length for type- $\vec{i}$ jobs is smaller than the number of type- $\vec{i}$ jobs that $M^{MW-E_{\vec{K}}}$ attempts to serve, i.e., $q_{\vec{i}} < M_{\vec{i}}^{MW-E_{\vec{K}}}(0)$. Therefore, the upper bound for $\lim_{t \downarrow 0} (III)/t$ is as follows:

$$\begin{aligned} \lim_{t \downarrow 0} \frac{1}{t} (III) &\leq \lim_{t \downarrow 0} \frac{1}{t} \mathbb{E} \left[\langle q(0), S(t) \rangle \mid q(0) = q, q \preceq M^{MW-E_{\vec{K}}}(0) \right] \\ &\leq \lim_{t \downarrow 0} \frac{1}{t} \sum_{\vec{i} \in \Gamma_{\vec{K}}} \mathbb{E} \left[M_{\vec{i}}^{MW-E_{\vec{K}}}(0) \cdot S_{\vec{i}}(t) \mid q(0) = q, q \preceq M^{MW-E_{\vec{K}}}(0) \right] \\ &\leq \lim_{t \downarrow 0} \frac{1}{t} \sum_{\vec{i} \in \Gamma_{\vec{K}}} (M_{\vec{i}}^{MW-E_{\vec{K}}}(0))^2 t \end{aligned}$$

$$\begin{aligned}
&= \left\| M^{MW-E_{\vec{K}}}(0) \right\|_2^2 \\
&\leq K_m^2,
\end{aligned}$$

where the last inequality is due to $\left\| M^{MW-E_{\vec{K}}}(0) \right\|_2^2 \leq \left\| M^{MW-E_{\vec{K}}}(0) \right\|_1^2 \leq K_m^2$.

By adding the above three limits for (I), (II), and (III) in (20), we have shown that $\Delta_M \circ L(q) \leq K_m + 2K_m^2 + \lambda - 2\delta\lambda_{\min} \|q\|_1$. Hence, for any $c > 0$, when

$$\|q\|_1 \geq \frac{K_m + 2K_m^2 + \lambda + c}{2\lambda_{\min}\delta}, \quad (27)$$

we can guarantee the drift $\Delta L(q) < -c$. Therefore, we can define the finite set C to contain all q for which (27) does not hold such that the Foster-Lyapunov condition is satisfied, and the CTMC $\{q(t)\}$ is positive recurrent. With the positive recurrence of the CTMC, we have proven that the empty state of the discretized system is positive recurrent. As a result, the empty state of the original continuous MRJ system is also positive recurrent. This concludes our stability proof. $\square$

We will use Lemma 1 in our proofs of Theorems 1 and 3 to 5 for the $\vec{K}$ -MW, X K -EMW, 2B K -EMW, and 2J $\vec{K}$ -EMW families, respectively. With the stability condition in Lemma 1 proven, we now give some extensions and related comments. First, we provide an equivalent sufficient stability condition for the discrete measure dominance condition (17).

Lemma 2 (Convex Hull Condition) *In the continuous MRJ setting, for a fixed $\vec{K} \in \mathbb{N}^d$, $\vec{K}$ -MW is stable if the $\vec{K}$ -arrival rate vector $\Lambda_{\vec{K}}$ is in the interior of the convex hull of the schedulable set $C_{\vec{K}}$. That is, the following convex hull condition holds*

$$\Lambda_{\vec{K}} \subseteq \text{int}(\text{ConvH}(C_{\vec{K}})). \quad (28)$$

Likewise, $\vec{K}$ -EMW is stable if the following convex hull condition holds for the efficient set $E_{\vec{K}}$

$$\Lambda_{\vec{K}} \subseteq \text{int}(\text{ConvH}(E_{\vec{K}})). \quad (29)$$

Specifically, the discrete measure dominance condition holds (17) for some service option distribution $\beta_{\vec{K}}$ over an efficient set $E_{\vec{K}}$ if and only if the convex hull condition (29) holds. Similarly, (17) holds for some $\beta_{\vec{K}}$ over $C_{\vec{K}}$ if and only if (28) holds.

Proof This lemma is a direct consequence of the definition of the convex hull. $\square$

Remark 1 (A Larger Efficient Set Maintains Stability). When $\vec{K}$ is fixed, for any two efficient sets $E_1 \subseteq E_2 \subseteq C_{\vec{K}}$, if $\Lambda_{\vec{K}} \subseteq \text{int}(\text{ConvH}(E_1))$, then clearly $\Lambda_{\vec{K}} \subseteq \text{int}(\text{ConvH}(E_2))$. Therefore, when the condition (17) holds for E_1 , both $\vec{K}$ -EMW with efficient set E_1 and with efficient set E_2 stabilize the system.

In addition to the $\vec{K}$ -MW and $\vec{K}$ -EMW policies, we also study $\vec{K}$ -MW-B and $\vec{K}$ -EMW-B, which add backfilling to use more of the capacity than the baseline policies. Empirically, $\vec{K}$ -MW-B and $\vec{K}$ -EMW-B achieve lower mean response time and a larger empirical stability region. We now prove that $\vec{K}$ -MW-B and $\vec{K}$ -EMW-B are stable under the same stability condition in Lemma 1, demonstrating that this empirical behavior is not a happenstance.

Lemma 3 (Backfilling Maintains Stability) *Under the same assumption of Lemma 1, if the discrete measure dominance (17) is satisfied by some service option distribution $\beta_{\vec{K}}$ over an efficient set $E_{\vec{K}}$, then the corresponding $\vec{K}$ -EMW-B is also stable. Likewise, when (17) is satisfied by $\beta_{\vec{K}}$ over $C_{\vec{K}}$, then $\vec{K}$ -MW-B is also stable.*

Proof In this proof, we focus on $\vec{K}$ -EMW-B, and the stability for $\vec{K}$ -MW-B follows from the case when the efficient set $E_{\vec{K}} = C_{\vec{K}}$.

When $\vec{K}$ is fixed, we define $\hat{S}(t)$ such that $\hat{S}_{\vec{i}}(t)$ denotes the number of type- $\vec{i}$ jobs actually served in the time period $(0, t]$. We decompose $\hat{S}_{\vec{i}}(t)$ such that $\hat{S}_{\vec{i}}(t) = \tilde{S}_{\vec{i}}(t) + B_{\vec{i}}(t)$, where $B_{\vec{i}}(t)$ denotes the additional number of type- $\vec{i}$ jobs served due to Backfilling, and $\tilde{S}_{\vec{i}}$ is as defined in Lemma 1 for $\vec{K}$ -EMW. Note that $\vec{K}$ -EMW-B is not a purely discretization-based policy, and we need to consider the original $\mathcal{S}$ -valued Markov process $\{x(t)\}_{t \geq 0}$ to determine which jobs are served by Backfilling. By explicitly letting $q : \mathcal{S} \rightarrow \mathbb{N}^{\Gamma_{\vec{K}}}$ such that $q_{\vec{i}}(x) = \sum_{\ell=1}^{\|x\|_0} \mathbb{1}\{x_\ell \in I_{\vec{i}}\}$, the Lyapunov function can still be chosen as before, but as a function of $x \in \mathcal{S}$. To be specific, we recall the Lyapunov function L of the countable state space $\mathbb{N}^{\Gamma_{\vec{K}}}$ (18) in the proof of Lemma 1, and likewise denote $\tilde{L}$ to be the Lyapunov function for the continuous state space $\mathcal{S}$, such that

$$\tilde{L}(x) := L(q(x)) = \|q(x)\|_2^2.$$

Then, when the current state is x and thus $q(x) \in \mathbb{N}^{\Gamma_{\vec{K}}}$, the drift $\Delta_B \circ \tilde{L}(x)$ under $\vec{K}$ -EMW-B is given by

$$\begin{aligned} \Delta_B \circ \tilde{L}(x) &:= \lim_{t \downarrow 0} \frac{1}{t} \mathbb{E} \left[\tilde{L}(q(x(t))) - \tilde{L}(q(x(0))) \mid x(0) = x \right] \\ &= \lim_{t \downarrow 0} \frac{1}{t} \mathbb{E} \left[\left\| q(x(0)) + A(t) - \hat{S}(t) \right\|_2^2 - \left\| q(x(0)) \right\|_2^2 \mid x(0) = x \right] \\ &= \lim_{t \downarrow 0} \frac{1}{t} \mathbb{E} \left[\left\| q(x(0)) + A(t) - \tilde{S}(t) - B(t) \right\|_2^2 - \left\| q(x(0)) \right\|_2^2 \mid x(0) = x \right]. \end{aligned}$$

Because $q(x(0)) + A(t) - \hat{S}(t) \succeq \vec{0}$ and $B(t) \succeq \vec{0}$ almost surely, we have that the following conditional expectation bound

$$\mathbb{E} \left[\left\| q(x(0)) + A(t) - \hat{S}(t) \right\|_2^2 \mid x(0) = x \right] \leq \mathbb{E} \left[\left\| q(0) + A(t) - \tilde{S}(t) \right\|_2^2 \mid q(0) = q(x) =: q \right].$$

Therefore, at any state $x \in \mathcal{S}$, the drift $\Delta_B \circ \tilde{L}(x)$ of $\tilde{L}$ under $\vec{K}$ -EMW-B is always upper bounded by the drift $\Delta_M \circ L(q(x))$ of $L(q(x))$ under $\vec{K}$ -EMW.

With the Lyapunov argument in the proof of Lemma 1, we conclude that the discrete measure dominance condition (17) is sufficient to apply the Foster-Lyapunov theorem [35], which guarantees Harris recurrence in this setting. To be specific, the Foster-Lyapunov condition is satisfied for any $c > 0$ and a corresponding compact set

$$C := \left\{ s \in \mathcal{S} : \|s\|_0 := \|q(s)\|_1 < \frac{K_m + 2K_m^2 + \lambda + c}{2\lambda_{\min}\delta} \right\}.$$

This argument shows the Harris recurrence of the set C . Note that all states in C have a bounded number of jobs, and the completion rate is always at least 1. As a result, all states in C can move to the empty state in bounded time with uniformly lower-bounded probability, demonstrating positive recurrence to the empty state. $\square$

In addition, by combining Lemma 3 with the proofs of Theorems 1 and 3 to 5 respectively, we can demonstrate the throughput-optimality of the $\vec{K}$ -MW-B, X $\vec{K}$ -EMW-B, 2B K -EMW-B, and 2J $\vec{K}$ -EMW-B families, under their respective assumptions.

Remark 2. In this paper, we primarily focus on the Backfilling technique that searches according to the arrival order, which is the Backfilling rule First-Fit uses. However, Lemma 3 holds for any Backfilling technique. For instance, in the MSJ setting ($d = 1$), the server can pack jobs by scanning the queue according to increasing or decreasing order of resource requirement, which are the Backfilling rule for Least-Server-First (LSF) or Best-Fit, respectively.

5.2 Proof of Theorem 1: Throughput-Optimality of the $\vec{K}$ -MW Family

The following lemma states that for any resource requirements random variable V with density f_V Lipschitz continuous, the discrete measure dominance condition (17) can be established with some appropriate discretization parameter $\vec{K}$ and a distribution $\beta_{\vec{K}}$ over $C_{\vec{K}}$. Our main result, Theorem 1, will then follow directly from combining Lemma 1 and Lemma 4.

Lemma 4 *In the continuous MRJ setting, assume that the job resource requirements random variable V has a Lipschitz continuous density f_V , and assume that given some $\lambda > 0$, there exists a policy π that stabilizes the queueing system. Then there exists some $\vec{K} \in \mathbb{N}^d$ and a distribution $\beta_{\vec{K}}$ over $C_{\vec{K}}$, such that the discretized service measure $\eta_{\vec{K}}^S$ satisfies the measure dominance (17).*

Proof We give the proof in the multi-resource (MRJ) setting, and the single-resource setting (MSJ) can be shown by setting $d = 1$.

Suppose that for a given $\lambda > 0$ and V , there exists some policy π (see Definition 3) that leads to a stable system. At the steady-state of this Markov process under π , we obtain a stationary measure β^π over $\pi(\mathcal{S})$, the set of service options used by π . The stability of π implies that $\eta^A = \eta^\pi$, where

$$\eta^\pi(dx) = \int_{\mathcal{S}} \sum_{r \in \pi(s)} \mathbb{1}\{r \in dx\} d\beta^\pi(s),$$

so that the rate of job arrivals with resource requirements in I equals the completion rate of jobs served with resource requirements in I , where I is any subset of $(0, 1]^d$.

In the steady state, because the empty state is positively recurrent, the Markov process assigns positive probability mass $\epsilon_0 > 0$ to the empty job state $\emptyset \in \mathcal{S}$. At the empty job state, the only service option is $\emptyset \in \mathcal{S} := \pi(\mathcal{S})$. We consider the conditional measure over $(0, 1]^d$

$$\eta^S = \mathcal{L}(\eta^\pi | \pi \neq \emptyset) = \frac{1}{1 - \epsilon_0} \eta^\pi,$$

and call η^S the *service measure*. The service measure is induced by $\{M_\alpha^\pi\}_{\alpha \in \mathcal{A}} = \pi(\mathcal{S}) \setminus \emptyset$, namely the family of continuous service options π uses. Note that the index set of these service options $\mathcal{A}$ is uncountable. With a probability measure $\mathcal{L}(\pi | \eta \neq \emptyset)$ over $\pi(\mathcal{S})$, we can

obtain a probability measure ζ over $\mathcal{A}$, which we call the reference distribution, and express the service measure as

$$\eta^S(dx) = \int_{\mathcal{A}} \sum_{r \in M_{\alpha}^{\pi}} \mathbb{1}\{r \in dx\} d\zeta(\alpha). \quad (30)$$

The above construction guarantees that η^A is absolutely continuous with respect to η^S , and that

$$\frac{d\eta^S}{d\eta^A}(x) \geq 1 + \epsilon, \quad \forall x \in \text{int}(\text{supp}(\eta^A)), \quad (31)$$

where $\epsilon = \frac{\epsilon_0}{1-\epsilon_0}$. We call this property “continuous measure dominance”.

Now our goal is to find some $\vec{K}$ such that $\vec{K}$ -MW stabilizes the system as well, by applying (17), with a service measure $\beta_{\vec{K}}$ over $C_{\vec{K}}$ chosen based on β^{π} . To be specific, we let $\vec{K} = (K, \dots, K)$ for some K to be determined later, and aim to build a $\vec{K}$ -discretized service measure $\eta_{\vec{K}}^S$ based on some distribution $\beta_{\vec{K}}$ over $C_{\vec{K}}$, such that there exists some $\delta > 0$ satisfying

$$\frac{\eta_{\vec{K}}^S(\vec{i})}{\eta_{\vec{K}}^A(\vec{i})} \geq 1 + \delta, \quad \text{for all } \vec{i} \in \Gamma_{\vec{K}}, \text{ for which } \eta_{\vec{K}}^A(\vec{i}) > 0, \quad (32)$$

where recall that for each job type $\vec{i}$, $\eta_{\vec{K}}^A(\vec{i}) := \eta^A(I_{\vec{i}})$.

In our construction of $\beta_{\vec{K}}$, we start by classifying each discretized job type $\vec{i} \in \Gamma_{\vec{K}}$ in either the default set $\mathcal{D}^{\vec{K}}$ or the exceptional set $\mathcal{E}^{\vec{K}}$. For any default index $\vec{i} \in \mathcal{D}^{\vec{K}}$, we rely on the continuous service options $\{M_{\alpha}^{\pi}\}_{\alpha \in \mathcal{A}}$ to propose corresponding $\vec{K}$ -discretized service options to serve type- $\vec{i}$ jobs, which we then modify. We will discuss the modification after we have defined the exceptional set. For any exceptional index $\vec{i} \in \mathcal{E}^{\vec{K}}$, we include the service option that serves a single type- $\vec{i}$ job. Specifically, we assign a total probability mass $\xi_{\vec{K}}$ of $\beta_{\vec{K}}$ to each service option $\{\vec{i}\}$ for $\vec{i} \in \mathcal{E}^{\vec{K}}$, such that

$$\beta_{\vec{K}}(\{\vec{i}\}) = (1 + \delta)\eta_{\vec{K}}^A(\{\vec{i}\}), \quad \text{for all } \vec{i} \in \mathcal{E}^{\vec{K}}. \quad (33)$$

We will then assign the remaining mass to the discrete service option family $\{M\}$ induced by the continuous service option family $\{M_{\alpha}^{\pi}\}$, where the probability will be given by (34) to control the default set $\mathcal{D}^{\vec{K}}$. We will show that as K becomes large, the arrival measure on $\mathcal{E}^{\vec{K}}$ vanishes, and for any job type in the set $\mathcal{D}^{\vec{K}}$, the gap between our service measure $\eta_{\vec{K}}^S(\vec{i})$ and $\eta^{\pi}(I_{\vec{i}})$ vanishes. This then allows us to prove discrete measure dominance for large enough $\vec{K}$.

We now introduce the exception set $\mathcal{E}^{\vec{K}}$, which is the union of three different subsets, namely the boundary set $\mathcal{B}^{\vec{K}}$, the empty-round-up set $\mathcal{T}^{\vec{K}}$, and the low-probability set $\mathcal{J}^{\vec{K}}(c_1, c_2)$, with parameters $c_1 > 0$ and $0 < c_2 < 1$. Then we guarantee discrete measure dominance for all $\vec{i} \in \mathcal{E}^{\vec{K}}$, by including in $\beta_{\vec{K}}$ the corresponding service option $\{\vec{i}\}$ and with mass $\beta_{\vec{K}}(\{\vec{i}\})$ given in (33). To bound the probability assigned to the service options $\{\vec{i}\}$, $\vec{i} \in \mathcal{E}^{\vec{K}}$, we need to upper bound the total arrival rate of each subset of $\mathcal{E}^{\vec{K}}$, and further show that the upper bound converges to zero. Specifically, we will show that $\sum_{\vec{i} \in \mathcal{E}^{\vec{K}}} \beta_{\vec{K}}(\{\vec{i}\}) = (1 + \delta)\eta_{\vec{K}}^A(\mathcal{E}^{\vec{K}}) = O(K^{-c_2})$. We note that δ can be arbitrarily chosen in $(0, \epsilon)$, which does not affect the existence of $\vec{K}$.

The first subset of $\mathcal{E}^{\vec{K}}$ is the boundary set $\mathcal{B}^{\vec{K}}$, which includes all boundary job types defined in (15), namely any job type that uses the entirety of any of the resources. Using the Lipschitz condition, we provide in Lemma 8 an upper bound on the maximum density $\|f\|_{\infty}$,

such that $\|f\|_\infty \leq L := L(C, d) < \infty$. The value of $L(C, d)$ is given by (C1), depending only on the Lipschitz constant C and the dimension d . We defer the proof to Appendix C. Therefore, we have that $\eta_{\vec{K}}^A(\mathcal{B}^{\vec{K}}) \leq \frac{L}{K^d} dK^{d-1} \leq LdK^{-1}$, because there are at most dK^{d-1} job types in $\mathcal{B}^{\vec{K}}$, and each type has arrival rate upper bounded by $\frac{L}{K^d}$. This completes our bound for $\mathcal{B}^{\vec{K}}$.

The next subset of $\mathcal{E}^{\vec{K}}$ is defined as the empty-round-up set $\mathcal{I}^{\vec{K}}$, which contains all non-boundary job types $\vec{i}$ with $\eta_{\vec{K}}^A(I_{\vec{i}}) > 0$ and $\eta_{\vec{K}}^A(I_{\vec{i}+\vec{1}}) = 0$. To bound the arrival measure on $\mathcal{I}^{\vec{K}}$, we consider the zero set $\mathcal{Z}^{\vec{K}} := \{\vec{i} \in \Gamma_{\vec{K}} : \eta_{\vec{K}}^A(\vec{i}) = 0\}$ of all job types, the corresponding buckets of which have zero arrival rate. Then the following inclusion holds:

$$\mathcal{I}^{\vec{K}} \subseteq \bigcup_{\vec{i} \in \mathcal{Z}^{\vec{K}}} \{\vec{i} - \vec{1}\}.$$

For any $\vec{i} \in \mathcal{I}^{\vec{K}}$, $f_V(\frac{\vec{i}+\vec{1}}{K}) = 0$. Because f_V is Lipschitz with constant C , $\sup_{v \in I_{\vec{i}}} f(v) \leq C\sqrt{d}/K$ for all $\vec{i} \in \mathcal{Z}^{\vec{K}}$. Therefore, $\eta_{\vec{K}}^A(\{\vec{i} - \vec{1}\}) \leq C\sqrt{d}K^{-D-1}$ for any $\vec{i} \in \mathcal{Z}^{\vec{K}}$, and consequently,

$$\eta_{\vec{K}}^A(\mathcal{I}^{\vec{K}}) \leq C\sqrt{d}K^{-D-1}K^D = C\sqrt{d}K^{-1},$$

where we use the total number of job types to bound the number of job types in $\mathcal{I}^{\vec{K}}$, i.e., $|\mathcal{I}^{\vec{K}}| \leq |\mathcal{Z}^{\vec{K}}| \leq |\Gamma_{\vec{K}}| = K^D$. This completes our bound for $\mathcal{I}^{\vec{K}}$.

The final subset of $\mathcal{E}^{\vec{K}}$ is the low-probability set $\mathcal{J}^{\vec{K}}(c_1, c_2)$ for some constants $c_1 > 0$, $0 < c_2 < 1$, such that

$$\mathcal{J}^{\vec{K}}(c_1, c_2) = \{\vec{i} \in \Gamma_{\vec{K}} : \eta_{\vec{K}}^A(\vec{i}) \leq c_1 K^{-(D+c_2)}\}.$$

The choice of the pair (c_1, c_2) does not affect the existence of $\vec{K}$, but affects the final value of $\vec{K}$. In this paper, we do not attempt to optimize $\vec{K}$. If concrete values of c_1, c_2 are needed, $c_1 = 1$ and $c_2 = 1/2$ suffice. The arrival measure on the set $\mathcal{J}^{\vec{K}}$ is then upper bounded by

$$c_1 K^{-c_2-D} |\mathcal{J}^{\vec{K}}(c_1, c_2)| \leq c_1 K^{-c_2-D} |\Gamma_{\vec{K}}| = c_1 K^{-c_2}.$$

This completes our bound for $\mathcal{J}^{\vec{K}}(c_1, c_2)$.

As mentioned before, for each exceptional job type $\vec{i} \in \mathcal{E}^{\vec{K}} := \mathcal{I}^{\vec{K}} \cup \mathcal{B}^{\vec{K}} \cup \mathcal{J}^{\vec{K}}(c_1, c_2)$, we include in $\beta_{\vec{K}}$ the service option $\{\vec{i}\}$, which only serves one type- $\vec{i}$ job, and let $\beta_{\vec{K}}(\{\vec{i}\}) = (1 + \delta)\eta_{\vec{K}}^A(\vec{i})$. This guarantees the discrete measure dominance holds for all $\vec{i} \in \mathcal{E}^{\vec{K}}$, i.e., $\eta_{\vec{K}}^S(\vec{i}) \geq (1 + \delta)\eta_{\vec{K}}^A(\vec{i})$. We also let $\xi_{\vec{K}} := \sum_{\vec{i} \in \mathcal{E}^{\vec{K}}} \beta_{\vec{K}}(\{\vec{i}\}) = (1 + \delta)\eta_{\vec{K}}^S(\mathcal{E}^{\vec{K}})$, which denotes the the summed probability mass $\beta_{\vec{K}}$ assigns to these options. The above discussion implies that

$$\begin{aligned} \xi_{\vec{K}} &= (1 + \delta)\eta_{\vec{K}}^A(\mathcal{I}^{\vec{K}} \cup \mathcal{B}^{\vec{K}} \cup \mathcal{J}^{\vec{K}}(c_1, c_2)) \\ &\leq (1 + \delta) \left[LdK^{-1} + C\sqrt{d}K^{-1} + c_1 K^{-c_2} \right] \\ &\leq (1 + \delta)(Ld + C\sqrt{d} + c_1)K^{-c_2}, \end{aligned}$$

and thus $\lim_{K \uparrow \infty} \xi_{\vec{K}} = 0$.

We now switch our focus to constructing $\beta_{\vec{K}}$ to achieve the discrete measure dominance for the default job types $\vec{i} \in \mathcal{D}^{\vec{K}} := \Gamma_{\vec{K}} \setminus \mathcal{E}^{\vec{K}}$. We consider the family of service options $\{M_\alpha^\pi\}_{\alpha \in \mathcal{A}}$ used by the continuous policy π . We will take advantage of $\{M_\alpha^\pi\}_{\alpha \in \mathcal{A}}$ to construct the discretized service options that we will use for $\beta_{\vec{K}}$.

We create a mapping from M_α^π to the discretized service options $M_\alpha^{\vec{K}}$. This mapping entry-wise rounds down each resource requirement r to the nearest smaller discretized bucket (job type). Formally speaking, given any $\alpha \in \mathcal{A}$, M_α^π contributes a density $\zeta^{\vec{K}}(\alpha) = (1 - \xi_{\vec{K}})\zeta(\alpha)$ to $M_\alpha^{\vec{K}} := \{\lfloor Kr \rfloor\}_{r \in M_\alpha^\pi}$. By using smaller job types, any valid M_α^π is mapped to a valid $M_\alpha^{\vec{K}}$ that does not exceed any resource capacity constraint. Then for any service option $M \in \{M_\alpha^{\vec{K}}\}_{\alpha \in \mathcal{A}}$,

$$\beta_{\vec{K}}(M) = \int_{\mathcal{A}} \mathbb{1}\{M = \{\lfloor Kr \rfloor\}_{r \in M_\alpha^\pi}\} d\zeta^{\vec{K}}(\alpha), \quad (34)$$

and we obtain a valid probability distribution $\beta_{\vec{K}}$, where the remaining $\mathcal{E}^{\vec{K}}$ indices were defined in (33).

The above construction of $\beta_{\vec{K}}$ means that, for each $\vec{i} \in \mathcal{D}^{\vec{K}}$, $\eta_{\vec{K}}^S(\vec{i})$ is given by

$$\eta_{\vec{K}}^S(\vec{i}) = \sum_{M \in \mathcal{C}_{\vec{K}}} M(\vec{i}) \beta_{\vec{K}}(M) = \int_{\mathcal{A}} \sum_{l=1}^{\|M_\alpha^\pi\|_0} \mathbb{1}\{M_\alpha^\pi(l) \in I_{\vec{i}+\vec{1}}\} d\zeta^{\vec{K}}(\alpha),$$

which can be lower bounded, due to (31) and the Lipschitz assumption $\|f\|_{Lip} = C$, as follows:

$$\begin{aligned} \eta_{\vec{K}}^S(\vec{i}) &= \int_{\mathcal{A}} \sum_{l=1}^{\|M_\alpha^\pi\|_0} \mathbb{1}\{M_\alpha^\pi(l) \in I_{\vec{i}+\vec{1}}\} (1 - \xi_{\vec{K}}) d\zeta(\alpha) \\ &= (1 - \xi_{\vec{K}}) \eta^S(I_{\vec{i}+\vec{1}}) \\ &\geq (1 + \epsilon)(1 - \xi_{\vec{K}}) \eta^A(I_{\vec{i}+\vec{1}}) \\ &= (1 + \epsilon)(1 - \xi_{\vec{K}}) \int_{I_{\vec{i}+\vec{1}}} \lambda f(x) dx \\ &\geq (1 + \epsilon)(1 - \xi_{\vec{K}}) \int_{I_{\vec{i}+\vec{1}}} \lambda \left(f(x - \frac{\vec{1}}{K}) - \frac{C\sqrt{d}}{K}\right) dx \\ &= (1 + \epsilon)(1 - \xi_{\vec{K}}) \left[\eta^A(\vec{i}) - \frac{\lambda C\sqrt{d}}{K^{d+1}} \right]. \end{aligned}$$

To conclude that $\eta_{\vec{K}}^S(\vec{i}) \geq (1 + \delta)\eta_{\vec{K}}^A(\vec{i})$, we need to show for any $\vec{i} \in \mathcal{D}^{\vec{K}}$,

$$\begin{aligned} (1 + \epsilon)(1 - \xi_{\vec{K}}) \left[\eta_{\vec{K}}^A(\vec{i}) - \frac{\lambda C\sqrt{d}}{K^{d+1}} \right] &\geq (1 + \delta)\eta_{\vec{K}}^A(\vec{i}), \\ \Leftrightarrow [(1 + \epsilon)(1 - \xi_{\vec{K}}) - (1 + \delta)] \eta_{\vec{K}}^A(\vec{i}) &\geq (1 + \epsilon)(1 - \xi_{\vec{K}}) \frac{\lambda C\sqrt{d}}{K^{d+1}}. \end{aligned}$$

Because $\vec{i} \notin \mathcal{J}^{\vec{K}}(c_1, c_2)$, $\eta_{\vec{K}}^A(\vec{i}) \geq c_1 K^{-(D+c_2)}$. It is enough to let K satisfy that

$$\begin{aligned} [(1 + \epsilon)(1 - \xi_{\vec{K}}) - (1 + \delta)] c_1 K^{-(d+c_2)} &\geq (1 + \epsilon)(1 - \xi_{\vec{K}}) \frac{\lambda C\sqrt{d}}{K^{d+1}}, \\ \Leftrightarrow [(1 + \epsilon)(1 - \xi_{\vec{K}}) - (1 + \delta)] c_1 &\geq (1 + \epsilon)(1 - \xi_{\vec{K}}) \lambda C\sqrt{d} K^{c_2-1}. \end{aligned} \quad (35)$$

When K goes to infinity, the limit of the L.H.S. of (35) is $c_1(\epsilon - \delta)$, and the limit of the R.H.S. of (35) is 0, because c_2 is chosen in $(0, 1)$. Therefore, the discrete measure dominance

holds also for $\vec{i} \in \mathcal{D}^{\vec{K}}$ when K is large enough. In particular, if we choose $c_1 = 1$ and $c_2 = \frac{1}{2}$, then it suffices to let

$$K \geq \left(\frac{(1+\delta)(1+\epsilon)(Ld + \frac{2+\delta}{1+\delta}C\sqrt{d} + 1)}{\epsilon - \delta} \right)^2.$$

Because δ can be arbitrarily small, we can let

$$K = \epsilon^{-2}[(1+\epsilon)(Ld + 2C\sqrt{d} + 1)]^2 = \Theta(\epsilon^{-2}).$$

□

From the above two lemmas, Theorem 1 follows immediately that the $\vec{K}$ -MW family is throughput-optimal for any resource requirement V with Lipschitz continuous p.d.f. f_V .

Theorem 1 (Throughput-optimality of $\vec{K}$ -MW) *In the continuous MRJ setting with exponential durations, for any resource requirement distribution V with Lipschitz continuous probability density function f_V over $(0, 1]^d$, the family of $\vec{K}$ -MW policies is throughput-optimal. That is, if the continuous MRJ system is stable under some policy π , then there exists some $\vec{K}$ such that $\vec{K}$ -MW stabilizes the system.*

Proof of Theorem 1 For any system that can be stabilized, with Lipschitz continuous f_V , Lemma 4 shows the existence of $\vec{K} = K\vec{1}$ and some service option distribution $\beta_{\vec{K}}$, such that the corresponding discretized service measure satisfies discrete measure dominance (17). This is a sufficient condition for the stability of $\vec{K}$ -MW, as proven in Lemma 1. □

5.3 Proof of Theorem 2: Throughput-Optimality of the $\vec{K}$ -nMSR Family

We next show Theorem 2, the throughput-optimality of the $\vec{K}$ -discretized nMSR ($\vec{K}$ -nMSR) family under the condition of Lipschitz continuous f_V . We first refer to one of the main results in Chen et al. [15], which is a stability condition for nonpreemptive Markovian Service Rate (nMSR).

Lemma 5 (Throughput-optimality of nMSR. Theorem 4 in [15], with notation modified.) *The class of nMSR policies is throughput-optimal. That is, if there exists a policy that stabilizes a multiresource-job system with J job types and arrival vector $\Lambda_J \in \mathbb{R}^J$, then there exists an nMSR policy, π , with $J^\pi \leq J$ working states that stabilizes the system. Specifically, an nMSR policy can stabilize a system with nonpreemptible jobs if and only if there are some $\epsilon > 0$ and*

$$(1+\epsilon)\Lambda_J \in \text{Conv}H(C^{(J)}), \quad (36)$$

where $C^{(J)}$ denotes the set of all possible schedules.

Now we prove Theorem 2, by combining Lemma 4 and Lemma 5.

Theorem 2 (Throughput-optimality of $\vec{K}$ -nMSR) *In the continuous MRJ setting, the family of $\vec{K}$ -nMSR policies is throughput-optimal for all V with Lipschitz continuous density f_V .*

Proof of Theorem 2 In the context of this work, we have $\Lambda_J = \Lambda_{\vec{K}}$, $J = \prod_{l=1}^d K_l$, and $C^{(J)} = C_{\vec{K}}$. In addition, Lemma 4 implies that when f_V is Lipschitz continuous, we can always find some $\vec{K}$ such that the discrete measure dominance condition (17) holds. The discrete measure dominance condition is equivalent to the convex hull condition (29), as shown in Lemma 2. This means the stability condition (36) of $\vec{K}$ -nMSR in Lemma 5 is satisfied. Therefore we conclude the stability of the CTMC $\{q_t\}_{t \geq 0}$ and the positive recurrence of the empty state. $\square$

5.4 Proof of Theorem 3: Throughput-Optimality of the Extreme-vertices $\vec{K}$ -EMW and $\vec{K}$ -EnMSR Families

Before we prove Theorem 3, we provide the following auxiliary lemma, which gives a necessary condition for a stable $\vec{K}$ -discretized policy.

Lemma 6 (Necessary Condition for Stable $\vec{K}$ -discretized Policy) *In the continuous MRJ setting, for a fixed $\vec{K} \in \mathbb{N}^d$, if there exists any $\vec{K}$ -discretized policy that stabilizes the system, then there exists a service option distribution $\beta_{\vec{K}}$ over $C_{\vec{K}}$ such that its corresponding service measure satisfies the discrete measure dominance (17).*

Proof Assume that the queueing system is stable under some $\vec{K}$ -discretized policy π , i.e., the empty state of the CTMP $\emptyset \in \mathcal{S}$ is positive recurrent. Then considering the CTMC $\{q(t)\}_{t \geq 0}$, we have that the empty state $\vec{0} := (0)_{\Gamma_{\vec{K}}}$ is also positively recurrent. Because the CTMC is irreducible, it is positive recurrent and thus ergodic. Hence, an ergodic distribution over the state space $\mathbb{N}^{\Gamma_{\vec{K}}}$ induces a stationary distribution $\beta'_{\vec{K}}$ over the set of all schedulable service options $\pi(\mathbb{N}^{\Gamma_{\vec{K}}}) \subset C_{\vec{K}}$, such that

1. The probability ϵ that the server serves no job is identical to the ergodic measure of the empty state $\vec{0}$, which is positive: $\epsilon = \beta'_{\vec{K}}(\vec{0}) > 0$.
2. The service rate vector under $\beta'_{\vec{K}}$ matches the arrival rate vector such that the equilibrium is achieved: $\mathbb{E}^{\beta'_{\vec{K}}} M = \Lambda_{\vec{K}}$.

Therefore, when we let $\beta_{\vec{K}}$ be the distribution of service options over $C_{\vec{K}}$ such that $\beta(M) = \frac{\beta'_{\vec{K}}(M)}{1-\epsilon}$ for all $M \neq \vec{0}$, the measure dominance (17) holds via the following inequality:

$$\begin{aligned} \sum_{M \in C_{\vec{K}}} \beta'_{\vec{K}}(M)M &= \sum_{M \in C_{\vec{K}} \setminus \vec{0}} \beta'_{\vec{K}}(M)M = (1-\epsilon) \sum_{M \in C_{\vec{K}} \setminus \vec{0}} \frac{\beta'_{\vec{K}}(M)}{1-\epsilon} M = \Lambda_{\vec{K}} \\ \Rightarrow \sum_{M \in C_{\vec{K}} \setminus \vec{0}} \beta_{\vec{K}}(M)M &= \sum_{M \in C_{\vec{K}} \setminus \vec{0}} \frac{\nu(M)}{1-\epsilon} M = \frac{1}{1-\epsilon} \Lambda_{\vec{K}} \succ \Lambda_{\vec{K}}. \end{aligned} \quad \square$$

Remark 3 (Sufficient and Necessary Condition for $\vec{K}$ -MW). Lemma 1 shows that the discrete measure dominance condition (17) is sufficient for the stability of $\vec{K}$ -MW. Lemma 2 shows that the convex hull condition (29) is an equivalent condition of (17), for the stability of $\vec{K}$ -MW. Lemma 6 shows either conditions (17) and (29) are also necessary.

Theorem 3, the throughput-optimality of the $X \vec{K}$ -EMW and $X \vec{K}$ -EnMSR families given Lipschitz continuous p.d.f. f_V of the resource requirement V , can be proven directly using Lemmas 1 and 4 to 6, and Remark 3.

Theorem 3 (Throughput-optimality of $X \vec{K}$ -EMW and $X \vec{K}$ -EnMSR) *In the continuous MRJ setting, given the same conditions as in Theorem 1, $X \vec{K}$ -EMW and $X \vec{K}$ -EnMSR are both throughput-optimal. Moreover, for any $\vec{K} \in \mathbb{N}^d$ fixed, $\vec{K}$ -MW, $X \vec{K}$ -EMW, $\vec{K}$ -nMSR, and $X \vec{K}$ -EnMSR are all stable for the same set of arrival rates.*

Proof of Theorem 3 Recall that E_K^X is the set of extreme vertices of $C_{\vec{K}}$. Therefore, E_K^X is the set of the extreme points of the polyhedron $\text{Conv}H(C_{\vec{K}})$, and $\text{Conv}H(C_{\vec{K}}) = \text{Conv}H(E_K^X)$.

By the convex hull condition (29) provided in Lemma 2 for the stability of the system, we conclude the sufficient and necessary condition for the stability of $X \vec{K}$ -EMW coincides with (17). Hence, $X \vec{K}$ -EMW and $\vec{K}$ -MW have identical stability regions. The throughput-optimality of $X \vec{K}$ -EMW is then implied directly from the throughput-optimality of $\vec{K}$ -MW by Theorem 1.

We note that the stability condition of $\vec{K}$ -nMSR (36) in Lemma 5 is equivalent to the stability condition (29). Also because $\text{Conv}H(C_{\vec{K}}) = \text{Conv}H(E_K^X)$, we conclude that $X \vec{K}$ -EnMSR has the same stability region as $\vec{K}$ -nMSR. The throughput-optimality of $X \vec{K}$ -EnMSR likewise holds. $\square$

Remark 4 (XP K -EMW and XP K -EMW-B). Recall that in Section 3.3.3, we introduced the Pairwise Extreme-vertices (XP) efficient set E_K^{XP} defined by (16). For any distinct $M_1, M_2 \in \tilde{C}_{K/2}$, the service option $M_1 + M_2$ can be written as a nontrivial convex combination of two distinct service options $2M_1, 2M_2 \in C_K$: $M_1 + M_2 = \frac{1}{2}(2M_1) + \frac{1}{2}(2M_2)$. Therefore, we guarantee that E_K^{XP} removes only service options that are not extreme points of C_K . Hence, $E_K^X \subset E_K^{\text{XP}} \subseteq \tilde{C}_K \subseteq C_K$, and $\text{Conv}H(E_K^{\text{XP}}) = \text{Conv}H(C_K)$. By Lemma 2, the XP K -EMW family is throughput-optimal in the MSJ setting when the resource requirement distribution has a Lipschitz continuous p.d.f. In addition, by Lemma 3, we also ensure the throughput-optimality of the XP K -EMW-B family under the above condition. Because no efficient algorithm is known for constructing E_K^X [33], E_K^{XP} provides a tractable proxy for E_K^X in practice. Empirically, the ratio $|E_K^{\text{XP}}|/|C_K|$ also converges to zero as K increases.

5.5 Necessary Stability Condition for an MRJ System

The following lemma provides an upper bound for the stability region of any policy. We will use this lemma to prove Theorems 4 and 5 in the Sections 5.6 and 5.7.

Lemma 7 *Consider an MRJ system with resource requirement distribution $V = (V_1, \dots, V_d)$. We let m be the resource type that has the largest mean requirement, i.e., $m := \arg \max_{1 \leq j \leq d} \mathbb{E}V_j$.*

Then the stability region for any policy is a subset of

$$\{\lambda > 0 : \lambda \mathbb{E}V_m < 1\}. \quad (37)$$

Proof We choose the test function $w_m : \mathcal{S} \rightarrow \mathbb{R}_{+,0}$ to be the total requirement of the type- m resource in the system, i.e., workload of type- m resource. Formally, when the state is given by $x = (\vec{v}_n)_{1 \leq n \leq N}$, $\phi(x) = \sum_{n=1}^N \vec{v}_n(m)$. Assume, for the sake of contradiction, that when $\lambda \geq \frac{1}{\mathbb{E}V_m}$, there exists a policy π stabilizing the system and resulting in the steady state distribution X . Then the ergodic probability at the empty state $\mathbb{P}(X = \emptyset) > 0$.

According to Glynn and Zeevi [36], at the steady state X , we have zero expected drift of $w_m(X)$ under the policy π :

$$\mathbb{E}[\Delta_\pi \circ w_m(X)] = 0. \quad (38)$$

Let H denote the stationary total resource requirement in service. As a result, the rate of decrease of $w_m(x)$ is H , where we recall that jobs have duration distribution $\text{Exp}(1)$. On the other hand, the rate of increase of ϕ due to arrivals is $\lambda \mathbb{E}V_m$. Therefore, we can express the expected drift explicitly, such that

$$\mathbb{E}[\Delta_\pi \circ w_m(X)] = \lambda \mathbb{E}V_m - \mathbb{E}[H|X \neq \emptyset] \mathbb{P}(X = \emptyset).$$

Because $H \leq 1$,

$$\mathbb{E}[\Delta_\pi \circ w_m(X)] > \lambda \mathbb{E}V_m - 1 \geq 0,$$

which contradicts the fact of zero expected drift (38). Therefore, we conclude that the stability region of any policy is a subset of $\{\lambda : \lambda \mathbb{E}V_m < 1\}$. $\square$

5.6 Proof of Theorem 4: Throughput-Optimality of the 2-Bucket K -EMW and $\vec{K}$ -EnMSR Families

We now prove Theorem 4, the throughput-optimality of the K -EMW and K -EnMSR families given f_V is weakly decreasing in $(0, 1]$.

Theorem 4 (2B K -EMW and 2B K -EnMSR) *In the continuous MSJ setting with the weakly decreasing density function of V , the 2B K -EMW and 2B K -EnMSR families are throughput-optimal. In particular, for any $\lambda < \frac{1}{\mathbb{E}V}$, there exists some K such that the queuing system is stable under 2B K -EMW and under 2B K -EnMSR. In particular, it is enough to let $K = 2^L$, where $L = \lfloor -\log_2(\frac{1}{\lambda} - \mathbb{E}V) \rfloor + 1$.*

Proof of Theorem 4 By Lemma 7, note that only if $\lambda < \frac{1}{\mathbb{E}V}$ can a system be stabilized by some policy. We then move to show that for any $\lambda < \frac{1}{\mathbb{E}V}$, we can always find some $K = 2^L$ such that 2B K -EMW is stable.

After we fix some integer L such that $K = 2^L$, we conduct a K -discretization as derived in (5). We recall that in Section 3.3.3, the service option is indexed by the type of the larger job it attempts to serve:

- For $k = 2^\ell \in [K]$ that is a power-of-two integer (including $k = 1$), we include the service option that serves $2^{L-\ell}$ type- k jobs simultaneously. Specifically, $M_{2^\ell} = \{(2^\ell, 2^{L-\ell})\}$.
- For $k \in [K]$ that is not a power-of-two integer, let $\ell := \ell(k) = \lceil \log_2 k \rceil$. We include the service option that serves $2^{L-\ell}$ type- k jobs and $2^{L-\ell}$ type- $(2^\ell - k)$ jobs simultaneously. Specifically, $M_k = \{(k, 2^{L-\ell}), (2^\ell - k, 2^{L-\ell})\}$.

Our remaining task is to assign appropriate probability mass $\beta_k := \beta(M_k)$ to each M_k , such that the corresponding discrete service measure η_K^S satisfies $\eta_K^S(k) > \eta_K^A(k)$ for all

$1 \leq k \leq K = 2^L$. As the stability conditions for 2B K -EMW and 2B K -EnMSR are the same, the discrete measure dominance condition (17) suffices to show both policies are stable.

To establish a service option distribution β_K , we first construct $L + 1$ sequences of nonnegative real numbers $\{p_j^{(\ell)}\}_{1 \leq j \leq 2^{L-\ell}}$, where sequences $\ell = 0, \dots, L$ are constructed iteratively. In round 0, $\{p_j^{(0)}\}_{1 \leq j \leq 2^L}$ is defined based on the requirement distribution vector $(p_k)_{1 \leq k \leq K}$, where $p_k := \mathbb{P}(V \in I_k)$. Then, in the r -th round, we construct $\{p_j^{(r)}\}_{1 \leq j \leq 2^{L-r}}$ recursively based on $\{p_j^{(r-1)}\}_{1 \leq j \leq 2^{L-r+1}}$ when $r \geq 1$. Intuitively, for each round r , the arrival probability type- j jobs with $j > 2^{L-r}$ are fully covered by service option M_k , and the sequence $p_j^{(r)}$ ($1 \leq j \leq 2^{L-r}$) represents the fraction of type- j jobs not covered by service option M_k , with $k > 2^{L-r}$. Additionally, in the r -th round, for each uncovered job type $j > 2^{L-r-1}$, we note that only M_j is able to serve some type- j jobs, among the remaining service options $\{M_k\}_{k \leq 2^{L-r}}$. Therefore, for each round $0 \leq r \leq L$ and each corresponding job type $2^{L-r-1} < j \leq 2^{L-r}$, we assign M_j some mass β_j proportional to $p_j^{(r)}$ to cover the remaining fraction of type- j jobs. These proxies of masses $p_j^{(r)}$ then form a sequence of non-negative numbers $\{p_k^{(L-\ell(k))}\}_{1 \leq k \leq K}$. After the construction of this a mass-proxy sequence $\{p_k^{(L-\ell(k))}\}_{1 \leq k \leq K}$, we will set the probability mass β_k assigned to M_k to be $\frac{(1+\epsilon)\lambda}{2^{L-\ell(k)}} p_k^{(L-\ell(k))}$, for some ϵ specified later. We will show that the discrete service measure condition induced by our service option distribution $\{\beta_k\}_{1 \leq k \leq K}$ is satisfied. Moreover, we will demonstrate that the masses β_k sum to at most 1 and hence the β_K is a valid probability distribution.

Now, we begin our 0-th round by simply letting $p_j^{(0)} = p_j$, for all $1 \leq j \leq L$. Because all masses p_j are nonnegative, $p_j^{(0)} \geq 0$ directly holds for each $2^{L-1} < j < 2^L$. We recall that we only take $\{p_j^{(0)}\}_{j > 2^{L-1}}$ into our mass-proxy sequence $\{p_k^{(L-\ell(k))}\}_{1 \leq k \leq K}$, but we will use the whole set $\{p_j^{(0)}\}_{1 \leq j \leq 2^L}$ in our following analysis.

In the 1-st round, we consider the uncovered job types $1 \leq j \leq 2^{L-1}$. We define $p_j^{(1)} := p_{2^L-j}^{(0)} - p_j^{(0)}$ be the difference of the complementary pair, where $1 \leq j < 2^{L-1}$, and define $p_{2^L-1}^{(1)} := p_{2^L-1}^{(0)}$ that copies its value in the previous round. Because $p_j^{(0)} = p_j$ is weakly decreasing with respect to its index j , we have the following inequality hold for each $1 \leq j \leq j' \leq 2^{L-1}$,

$$p_{2^L-j}^{(0)} \geq p_{2^L-j'}^{(0)} \geq p_{j'}^{(0)} \geq p_j^{(0)},$$

which implies $p_j^{(1)} \geq p_{j'}^{(1)} \geq 0$. Therefore, we have that $p_j^{(1)}$ is also weakly decreasing. Thereafter, in round r , the weakly decreasing property does not include the boundary index 2^{L-r} , because we do not need it to subtract an element in the next round but just copy it, so that its non-negativity always holds.

We continue the above iterations, and in the r -th round, we have a nonnegative sequence $\{p_j^{(r-1)}\}_{j \leq 2^{L-r+1}}$ obtained in the previous round, of which $\{p_j^{(r-1)}\}_{j < 2^{L-r+1}}$ is weakly decreasing. By defining $p_j^{(r)} := p_{2^{L-r+1}-j}^{(r-1)} - p_j^{(r-1)}$ for $1 \leq j < 2^{L-r}$, and copying $p_{2^{L-r}}^{(r)} := p_{2^{L-r}}^{(r-1)}$, we obtain a new sequence $\{p_j^{(r)}\}_{1 \leq j \leq 2^{L-r}}$. This sequence is weakly decreasing (excluding the boundary element $p_{2^{L-r}}^{(r)}$) and nonnegative, due to the weakly decreasing property of $\{p_j^{(r-1)}\}_{j < 2^{L-r+1}}$. Finally we reach the $(L-1)$ -th round and have only $p_2^{(L-2)}$ and $p_1^{(L-2)}$ from the $(L-2)$ -th round. We notice that there is no complementary index pair,

and therefore define $p_2^{(L-1)} := p_2^{(L-1)}$ and directly move to the L -th round, where it remains only to define $p_1^{(L)} := p_1^{(L-2)}$.

Note that our sequence $\{p_j^{(L-\ell(j))}\}_{1 \leq j \leq K}$ use only K elements in $\{p_j^{(r)}\}_{1 \leq j \leq 2^{L-r}, 0 \leq r \leq L}$. We let $\epsilon > 0$ be a real number whose value will be specified later, and assign the probability mass $\beta_k := \frac{(1+\epsilon)\lambda}{2^{L-\ell(k)}} p_k^{(L-\ell(k))}$ to the service option M_k .

Before we show that $\{\beta_k\}_{k=1}^K$ is a valid probability measure, we prove that the discrete measure dominance (17) holds. For each non-power-of-two integer k , the service option that attempt to serve some type- k job(s) are $M_{2^\ell-k}$, $\ell > \ell(k)$, and M_k . Service option $M_{2^\ell-k}$ attempts to serve $2^{L-\ell}$ type- k jobs and service option M_k attempts to serve $2^{L-\ell(k)}$ jobs. Therefore, the discrete service measure $\eta_K^S(k)$ satisfies that

$$\begin{aligned}
\eta_K^S(k) &= 2^{L-\ell(k)} \beta_k + \sum_{\ell=\ell(k)}^L 2^{L-\ell} \beta_{2^\ell-k} \\
&= 2^{L-\ell(k)} \frac{(1+\epsilon)\lambda}{2^{L-\ell(k)}} p_k^{(L-\ell(k))} + \sum_{\ell=\ell(k)}^L 2^{L-\ell} \frac{(1+\epsilon)\lambda}{2^{L-\ell}} p_{2^\ell-k}^{(L-\ell)} \\
&= (1+\epsilon)\lambda \left[p_k^{(L-\ell(k))} + \sum_{\ell=\ell(k)}^L p_{2^\ell-k}^{(L-\ell)} \right] \\
&= (1+\epsilon)\lambda p_k \\
&> \eta_K^A(k),
\end{aligned} \tag{39}$$

where (39) is due to the iterative decomposition $p_j^{(r)} = p_{2^{L-r+1}-j}^{(r-1)} - p_j^{(r-1)}$. For each power-of-two index k , the service option that attempt to serve some type- k job(s) are $M_{2^\ell-k}$, $\ell \geq \ell(k) + 1$, and M_k . Likewise, using the fact that $p_k^{(L-\ell(k))} := p_k^{(L-\ell(k)+1)}$ in addition to the above argument, we obtain that $\eta_K^S(k) = (1+\epsilon)\lambda p_k > \eta_K^A(k)$, and conclude discrete measure dominance.

We next show that there exists L, ϵ , such that β_K is a valid probability distribution, by arguing that $\sum_{1 \leq j \leq 2^L} \beta_j \leq (1+\epsilon)\lambda \mathbb{E}V + o(2^{-L})$. To achieve this, we first claim that the expectation of V can be expressed as

$$\mathbb{E}V = \sum_{j=1}^{2^L} \left[\frac{\beta_j}{(1+\epsilon)\lambda} \sum_{k=1}^{2^L} M_j^{(k)} \mathbb{E}[V|V \in I_k] \right]. \tag{40}$$

Equation (40) holds because

$$\frac{\beta_j}{(1+\epsilon)\lambda} = 2^{-L+\ell(j)} p_j^{(L-\ell(j))},$$

and $M_j^{(k)} = 2^{L-\ell(j)}$ if and only if M_j serves at least one jobs in the bucket k . Therefore, the R.H.S. of (40) can be written as

$$\begin{aligned}
& \sum_{j=1}^{2^L} \left[p_j^{(L-\ell(j))} \sum_{k=1}^{2^L} \mathbb{1}\{M_j^{(k)} > 0\} \mathbb{E}[V|V \in I_k] \right] \\
&= \sum_{k=1}^{2^L} \left[\left(\sum_{j=1}^{2^L} p_j^{(L-\ell(j))} \mathbb{1}\{M_j^{(k)} > 0\} \right) \mathbb{E}[V|V \in I_k] \right] \\
&= \sum_{k=1}^{2^L} (p_k \mathbb{E}[V|V \in I_k]) \\
&= \mathbb{E}V,
\end{aligned} \tag{41}$$

where step (41) is due to the iterative decomposition $p_j^{(r)} = p_{2^{L-r+1}-j}^{(r-1)} - p_j^{(r-1)}$ that we also applied to justify (39), or alternatively, because the summed serving rates at interval I_k is $(1+\epsilon)\lambda p_k$:

$$\sum_{j=1}^{2^L} p_j^{(L-\ell(j))} \mathbb{1}\{M_j^{(k)} > 0\} = p^{(k)}. \tag{42}$$

After decomposing $\mathbb{E}V$, we likewise decompose the summed mass $\sum_{1 \leq j \leq 2^L} \frac{\beta_j}{(1+\epsilon)\lambda}$. We use the property that for any j , the service option M_j does not waste any capacity, i.e., $\sum_{k=1}^{2^L} M_j^{(k)} \frac{k}{2^L} = 1$. Therefore,

$$\sum_{j=1}^{2^L} \left[\frac{\beta_j}{(1+\epsilon)\lambda} \left(\sum_{k=1}^{2^L} M_j^{(k)} \frac{k}{2^L} \right) \right] = \sum_{j=1}^{2^L} \frac{\beta_j}{(1+\epsilon)\lambda}.$$

In addition, $\frac{k-1}{2^L} \leq \mathbb{E}[V|V \in I_k] \leq \frac{k}{2^L}$ means the waste of capacity by up-rounding each job's requirement is controlled: $\frac{k}{2^L} - \mathbb{E}[V|V \in I_k] \leq 2^{-L}$. This result leads to the following bound:

$$\begin{aligned}
& \sum_{j=1}^{2^L} \frac{\beta_j}{(1+\epsilon)\lambda} - \mathbb{E}V \\
&= \sum_{j=1}^{2^L} \left[\frac{\beta_j}{(1+\epsilon)\lambda} \sum_{k=1}^{2^L} \left[M_j^{(k)} \left(\frac{k}{2^L} - \mathbb{E}[V|V \in I_k] \right) \right] \right] \\
&\leq \sum_{j=1}^{2^L} \left[\frac{\beta_j}{(1+\epsilon)\lambda} \sum_{k=1}^{2^L} \left(M_j^{(k)} \frac{1}{2^L} \right) \right] \\
&= \frac{1}{2^L} \sum_{j=1}^{2^L} \left(\sum_{k=1}^{2^L} \mathbb{1}\{M_j^{(k)} > 0\} p_j^{(L-\ell(j))} \right) \\
&= \frac{1}{2^L},
\end{aligned} \tag{43}$$

where step (43) is due to the decomposition of $\mathbb{E}V$ (40), and step (44) is again due to (42).

It now remains to bound the summed mass. By letting $L = \lceil -\log_2(\frac{1}{(1+\epsilon)\lambda} - \mathbb{E}V) \rceil$ (and thus $K = 2^L = \Theta(\frac{1}{1-\lambda\mathbb{E}V})$), we have

$$\sum_{j=1}^{2^L} \beta_j \leq (1+\epsilon)\lambda\mathbb{E}V + \frac{(1+\epsilon)\lambda}{2^L} \leq 1.$$

As ϵ can be chosen arbitrarily small, we can guarantee a valid service option distribution β_K with $L = \lfloor -\log_2(\frac{1}{\lambda} - \mathbb{E}V) \rfloor + 1$. Because we have shown that this service option distribution leads to the discrete measure dominance, we finally conclude Theorem 4. $\square$

5.7 Proof of Theorem 5: Throughput-Optimality of the 2-Job $\vec{K}$ -EMW and $\vec{K}$ -EnMSR Families

We now prove Theorem 5, the throughput-optimality of the 2J $\vec{K}$ -EMW and 2J $\vec{K}$ -EnMSR families when the p.d.f. f_V of the resource requirement distribution V is centrally symmetric with respect to $\frac{1}{2}\vec{1}$ and Lipschitz continuous.

Theorem 5 (2J $\vec{K}$ -EMW and 2J $\vec{K}$ -EnMSR) *In the continuous MRJ setting, assume that the p.d.f. f_V is Lipschitz continuous and centrally symmetric with respect to $\frac{1}{2}\vec{1}$. In this case, both the 2J $\vec{K}$ -EMW and 2J $\vec{K}$ -EnMSR families are throughput-optimal. To be specific, for any arrival rate $\lambda < 2$, there exists some $\vec{K}$ such that both 2J $\vec{K}$ -EMW and 2J $\vec{K}$ -EnMSR stabilize the system.*

Moreover, if $V \sim \text{Unif}((0, 1]^d)$, any $\vec{K} \succeq K\vec{1}$ stabilizes the system, where K is the smallest odd integer such that $K \geq \lfloor \frac{2\lambda d}{2-\lambda} \rfloor + 1$. Additionally, in the single-resource setting, it suffices to choose an odd K such that $K \geq \lfloor \frac{\lambda}{2-\lambda} \rfloor + 1$.

Proof of Theorem 5 By Lemma 7, we note that any system that can be stabilized should satisfy $\lambda < 2$. We next prove that for any $\lambda < 2$, we can find some $\vec{K}$ such that our 2J $\vec{K}$ -EMW stabilizes the system. Our goal is to construct a service option distribution $\beta_{\vec{K}}$ satisfying discrete measure dominance (17) for some $\vec{K}$, and use Lemma 1 to demonstrate the stability given any $\lambda < 2$. We start by considering the general Lipschitz and centrally symmetric case, then we prove the special case result when V is uniform.

Our proof for the general case follows a similar breakdown into the exceptional and default sets as Lemma 4, but our only exceptional job types are the boundary job types in $\mathcal{B}^{\vec{K}}$.

We consider $\vec{K} = K\vec{1}$ for some odd K . Recall in Section 3.3.3 that we label the service option that serves only one boundary-type job $M_{\vec{i}}$, if the job is in the boundary set $\vec{i} \in \mathcal{B}^{\vec{K}}$, i.e., $M_{\vec{i}} = \{\vec{i}\}$. For $\vec{i} \notin \mathcal{B}^{\vec{K}}$, $M_{\vec{i}}$ has a different meaning. Apart from the boundary set, we label the service option that serves both one type- $\vec{i}$ job and one type- $(\vec{K} - \vec{i})$ job simultaneously $M_{\vec{i}}$, for each $\vec{i}$ such that $1 \leq \vec{i}_1 \leq \frac{K-1}{2}$ and $1 \leq \vec{i}_m \leq K-1$, where $2 \leq m \leq d$. Formally speaking, we let $\mathcal{H}^{\vec{K}} := [\frac{K-1}{2}] \times [K-1] \times \dots \times [K-1]$ denote the set of all job types with type-1 resource requirement at most $\frac{K-1}{2}$, and let $M_{\vec{i}} = \{\vec{i}, \vec{K} - \vec{i}\}$ for each $\vec{i} \in \mathcal{H}^{\vec{K}}$.

For simplicity, we let $\beta_{\vec{i}} := \beta_{\vec{K}}(M_{\vec{i}})$ for service option $M_{\vec{i}}$. We fix some positive ϵ and begin by assigning the mass $\beta_{\vec{i}} = (1+\epsilon)\eta_{\vec{K}}^A(\vec{i})$ to the policy $M_{\vec{i}}$ for each $\vec{i} \in \mathcal{B}^{\vec{K}}$, and assign the mass $\beta_{\vec{i}} = (1+\epsilon)\max(\eta_{\vec{K}}^A(\vec{i}), \eta^A(\vec{K} - \vec{i}))$ for each $\vec{i} \in \mathcal{H}^{\vec{K}}$. Then the discrete measure

dominance (17) is satisfied. It remains to show $\beta_{\vec{K}}$ is a valid probability distribution, i.e., $\sum_{\vec{i} \in \mathcal{B}^{\vec{K}} \cup \mathcal{H}^{\vec{K}}} \beta_{\vec{i}} \leq 1$. We apply Lemma 8 using the Lipschitz constant C of f_V , so we have $\sup f_V \leq L := L(C, d)$, and can bound $\eta_{\vec{K}}^S(\mathcal{B}^{\vec{K}}) = \sum_{\vec{i} \in \mathcal{B}^{\vec{K}}} \beta_{\vec{i}}$ such that

$$\eta_{\vec{K}}^S(\mathcal{B}^{\vec{K}}) \leq LK^{-d}dK^{d-1} = LdK^{-1}.$$

Then, for each non-boundary job type $\vec{i} \in \mathcal{H}^{\vec{K}}$, we bound the difference in service measure between the two job types served by $M_{\vec{i}}$.

$$\begin{aligned} |\eta_{\vec{K}}^A(\vec{i}) - \eta_{\vec{K}}^A(\vec{K} - \vec{i})| &= \lambda \left| \int_{I_{\vec{i}}} - \int_{I_{\vec{K} - \vec{i}}} \right| f_V(x) dx \\ &= \lambda \left| \int_{I_{\vec{i}}} f_V(x) dx - \int_{I_{\vec{K} - \vec{i}}} f_V(\vec{1} - x) dx \right| \\ &= \lambda \left| \int_{I_{\vec{i}}} (f_V(x) - f_V(x + K^{-1}\vec{1})) dx \right| \\ &\leq \lambda \int_{I_{\vec{i}}} |f_V(x) - f_V(x + K^{-1}\vec{1})| dx \\ &\leq \lambda \int_{I_{\vec{i}}} C\sqrt{d}K^{-1} dx \\ &= \lambda C\sqrt{d}K^{-d-1}, \end{aligned}$$

As a result, we have $\beta_{\vec{i}} \leq (1 + \epsilon)[\eta_{\vec{K}}^A(\vec{i}) + C\sqrt{d}K^{-d-1}]$. Therefore, because the number of non-boundary job types $|\mathcal{H}^{\vec{K}}| = \frac{1}{2}(\frac{K-1}{K})^d$, the summed mass is bounded by

$$\begin{aligned} \sum_{\vec{i} \in \mathcal{B}^{\vec{K}} \cup \mathcal{H}^{\vec{K}}} \beta_{\vec{i}} &\leq (1 + \epsilon)\lambda \left[LdK^{-1} + \sum_{\vec{i} \in \mathcal{H}^{\vec{K}}} \left(\frac{1}{\lambda} \eta_{\vec{K}}^A(\vec{i}) + C\sqrt{d}K^{-d-1} \right) \right] \\ &\leq (1 + \epsilon)\lambda \left[\left(Ld + \frac{C\sqrt{d}}{2} \left(\frac{K-1}{K} \right)^d \right) K^{-1} + \frac{1}{\lambda} \sum_{\vec{i} \in \mathcal{H}^{\vec{K}}} \eta_{\vec{K}}^A(\vec{i}) \right] \\ &< (1 + \epsilon)\lambda \left[\left(Ld + \frac{C\sqrt{d}}{2} \right) K^{-1} + \frac{1}{2} \right]. \end{aligned} \tag{45}$$

Here, the last inequality (45) holds because the two disjoint sets $\mathcal{H}^{\vec{K}}$ and $\vec{K} - \mathcal{H}^{\vec{K}}$ have identical arrival measure, and their union is a subset of $\Gamma_{\vec{K}}$, which has total arrival measure at most λ . As a result, $\sum_{\vec{i} \in \mathcal{H}^{\vec{K}}} \eta_{\vec{K}}^A(\vec{i}) \leq \frac{1}{2}$.

Using the upper bound (45), we choose K to be such that

$$\begin{aligned} (1 + \epsilon)\lambda \left[\left(Ld + \frac{C\sqrt{d}}{2} \right) K^{-1} + \frac{1}{2} \right] &\leq 1 \\ \Leftrightarrow K &\geq \left(\frac{1}{(1 + \epsilon)\lambda} - \frac{1}{2} \right)^{-1} \left(Ld + \frac{C\sqrt{d}}{2} \right). \end{aligned}$$

This completes the general centrally symmetric case. We now discuss a few special-case results. For the multi-resource uniform setting, $L = 1$ and $C = 0$. Thus it suffices to let $K \geq \lceil \frac{2d\lambda(1+\epsilon)}{2-(1+\epsilon)\lambda} \rceil$. By letting ϵ be small enough, we note that $K = \lfloor \frac{2\lambda d}{2-\lambda} \rfloor + 1$ enables discrete measure dominance (17) to hold.

We then show stability in the uniform single-resource (MSJ) setting. With $M_K = \{K\}$, $M_1 = \{1, K-1\}, \dots, M_{\frac{K-1}{2}} := \{\frac{K-1}{2}, \frac{K+1}{2}\}$, we let $\beta_K = \beta_1 = \dots, \beta_{\frac{K-1}{2}} = (1+\epsilon)\lambda/K$ for some $\epsilon > 0$, so that discrete measure dominance (17) can be established. It remains to ensure

$$\beta_K + \sum_{j=1}^{\frac{K-1}{2}} \beta_j = (1+\epsilon)\lambda \frac{K+1}{2K} \leq 1 \Rightarrow K \geq \lceil \frac{(1+\epsilon)\lambda}{2-(1+\epsilon)\lambda} \rceil.$$

Because ϵ can be arbitrarily small, it is enough to let $K = \lfloor \frac{\lambda}{2-\lambda} \rfloor + 1$.

Finally, by applying Lemmas 1 and 5, the stability of the system is achieved for any $\lambda < 2$, respectively under 2J $\tilde{K}$ -EMW and 2J $\tilde{K}$ -EnMSR. $\square$

6 Empirical Results

In this section, we use simulations to validate our main results for both parametric distributions and Google Borg trace data [16]. We compare our policies with several index-based benchmarks: FCFS, First-Fit, Best-Fit, and Least-Server-First (LSF). These index-based policies can also be naturally extended to the continuous-resource setting. The detailed definitions of First-Fit, Best-Fit, and LSF are given in Appendix B. In particular, we note that LSF is equivalent to naive MaxWeight in the continuous MSJ setting, and very similar in the Google Borg setting. When job resource requirements are continuous and almost surely distinct, each “job type” contains at most one job. As a result, naive MaxWeight always selects the feasible set containing the largest number of jobs, which is equivalent to batching jobs in increasing order of their resource requirements.

In Section 6.1, we simulate MSJ systems with i.i.d. parametric continuous resource requirements, to verify the throughput-optimality of our theoretically-motivated policies. In addition, we demonstrate that with a large discretization parameter K , the mean response time of K -EMW-B policies is consistently better than all prior index-based policies.

In Section 6.2, we compare our policy with the prior index-based and class-based policies, using resource requirement demand from a Google Borg trace. In this practical setting, although the requirements are not continuous or i.i.d., the performance advantages of our policies generalize beyond our model.

All data and code in this section can be found in the following repository: <https://github.com/isaacg1/buckets>.

6.1 Comparison of Mean Response Time under Parametric Requirement Distribution

We first compare the theoretically-motivated family of policies with index-based policies (FCFS, First-Fit, Best-Fit, and LSF), where the job resource requirements are sampled i.i.d. from some one-dimensional parametric continuous distributions.

We simulate the system with 10^6 job arrivals. In the simulation, if the queue length ever exceeds 10^4 , or the mean response time exceeds 10^3 , which we represent using dotted lines in our plot, we consider the system to be indistinguishable from unstable via simulation.

Given a known resource requirement distribution V , let D_V denote the theoretical stability region, defined as the set of all arrival rates $\lambda > 0$ such that the system is stable under some policy with arrival rate λ and distribution V . We further define $\lambda_V^* := \sup D_V$ as the stability boundary of D_V . In the following experiments, we use the mean response time $\mathbb{E}[T(\lambda)]$ as the performance metric, and visualize them against the system load

$$\rho := \frac{\lambda}{\lambda_V^*}.$$

The resource distributions we use in our simulation are uniform (Section 6.1.1), truncated normal (Section 6.1.2), bounded Lomax and triangular (Section 6.1.3), and symmetric triangular (Section 6.1.4), for which the p.d.f. are constant, centrally symmetric, weakly decreasing, and general Lipschitz, respectively.

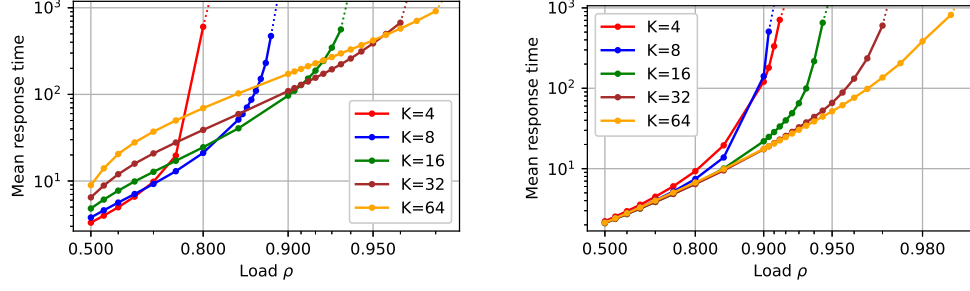
6.1.1 Uniformly Distributed Resource Requirement

We first consider the case where job resource requirements follow a uniform distribution on $(0, 1]$. Our primary focus is on the 2-Job K -discretized Efficient MaxWeight (2J K -EMW) policy, introduced in Section 3.3.3, which we proved to achieve throughput-optimality in Theorem 5. In addition, we compare our theoretically-motivated policies with index-based policies, to demonstrate their empirical advantages in reducing the mean response time. Note that the stability boundary is $\lambda_V^* = 2$.

In Fig. 1a, we compare the mean response times under 2J K -EMW, with $K = 4, 8, 16, 32$, and 64 . Because the policy may leave additional space available, Fig. 1b further compares the performance of 2J K -EMW with Backfilling (2J K -EMW-B), using the same set of discretization parameters K .

In Fig. 1a, we observe that for each K , the mean response time of 2J K -EMW increases when the load ρ grows, and jumps to indistinguishable from unstable. From now on, we call the set of loads for which the policy is demonstrated to be stable the “empirically-stable” region, which is an empirical lower bound of the theoretical stability region. As K increases, the boundary of the empirically-stable region increases towards 1, which verifies Theorem 5. In Fig. 1b, the empirically-stable region of 2J K -EMW-B also approaches to $[0, 1)$, which is similar to Fig. 1a.

In Fig. 1a, larger values of K result in a worse mean response time until the arrival rate becomes close to the stability boundary. In contrast, in Fig. 1b, when Backfilling is applied, a larger discretization parameter K generally leads to a smaller mean response time. Moreover, comparing the two sets of plots, it is clear that Backfilling consistently yields a smaller mean response time. We conjecture that this behavior arises because when the load is low and K is large, the system may not always have two complementary jobs. Hence, 2J K -EMW may be unable to pack two jobs for joint service, which wastes some capacity. In contrast, Backfilling mitigates this issue, allowing the server to pack non-complementary jobs and possibly additional small jobs. Therefore, in practice, we recommend choosing a larger enough K to ensure stability, as derived in Theorem 5, but not much larger, and applying Backfilling. As a result, the corresponding policy can achieve stability, computational efficiency, and a better mean response time.



(a) Mean response times under 2J K -EMW. (b) Mean response times under 2J K -EMW with Backfilling.

Fig. 1: Log-log plots of mean response times as functions of load ρ , under 2J K -EMW with or without Backfilling, where $K = 4, 8, 16, 32, 64$ and $V \sim \text{Uniform}((0, 1])$. Simulation duration is 10^6 jobs.

In Fig. 2, we compare 2J 64-EMW-B, the best policy in Fig. 1b, with index-based policies (FCFS, First-Fit, Best-Fit, and LSF). The simulation results show that naive FCFS, Best-Fit, and LSF have small empirically-stable regions, whereas First-Fit achieves a stability boundary close to $\rho = 1$. Among the policies compared in Fig. 2, 2J 64-EMW-B achieves the largest empirically-stable region and the least mean response time under high load.

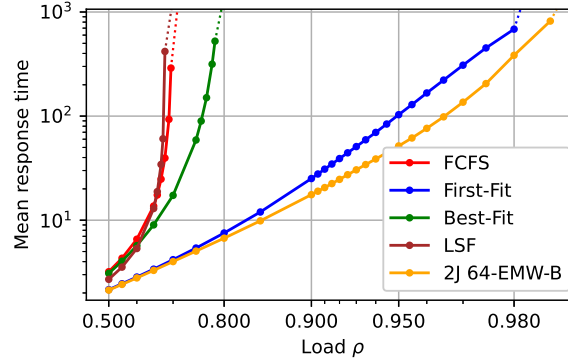


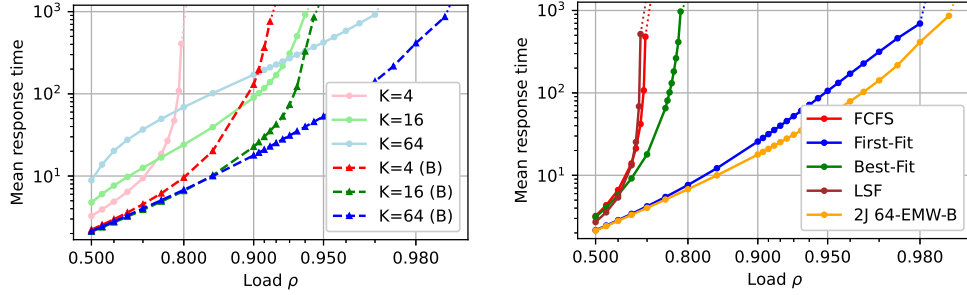
Fig. 2: Log-log plot of mean response times v.s. load ρ , under theoretically-motivated and index-based policies when $V \sim \text{Uniform}((0, 1])$. Simulation duration is 10^6 jobs.

6.1.2 Centrally Symmetric (Truncated Normal) Resource Requirement

We next consider a symmetric but non-uniform requirement distribution, to strengthen our empirical results in a setting of general symmetry. We further compare 2J K -EMW-B against K -MW-B, to show that choosing an appropriate efficient set helps to achieve both computational efficiency and a better mean response time.

The job resource requirement V follows a truncated normal distribution $T\mathcal{N}(0.5, 1)$ supported on $[0, 1]$, i.e., $\mathcal{L}(V) = \mathcal{L}(Z|Z \in [0, 1])$, where $Z \sim \mathcal{N}(0.5, 1)$. Because this distribution is symmetric with mean 0.5, we have $\lambda_V^* = 2$, and we know that the 2-Job K -Efficient MaxWeight (2J K -EMW) family is throughput-optimal, according to Theorem 5.

Our observations from Section 6.1.1 continue to hold in this centrally symmetric setting: In the absence of Backfilling, larger K leads to worse mean response times until the arrival rate approaches the stability boundary, whereas larger K combined with Backfilling results in uniformly better mean response times.



(a) Mean response times under 2J K -EMW and 2J K -EMW-B. The curves for 2J K -EMW-B have labels ending with (B). (b) Mean response times under our theoretically-motivated policy and index-based policies.

Fig. 3: Log-log plots for mean response times v.s. load ρ under 2J K -EMW with or without Backfilling, where $K = 4, 16, 64$ (left), and under index-based policies and the selected theoretically-motivated policy 2J 64-EMW-B (right). $V \sim T\mathcal{N}(0.5, 1)$. Simulation duration is 10^6 jobs.

Next, in Fig. 3b, we compare index-based policies with our theoretically-motivated policies. Consistent with previous results, 2J 64-EMW-B achieves the largest empirically-stable regions, whereas FCFS, Best-Fit, LSF are indistinguishable from unstable under relatively small loads ($\rho \leq 0.8$).

First-Fit, though leading to a worse mean response time than K -EMW-B, remains stable under high load, and we conjecture that it is stable in the single-resource setting when the resource requirement distribution is centrally symmetric. The reason for our conjecture is that First-Fit is proven to be throughput-optimal in Coffman and Stolyar [4], in a setting close to our central symmetry setting. To be specific, [4] considers the

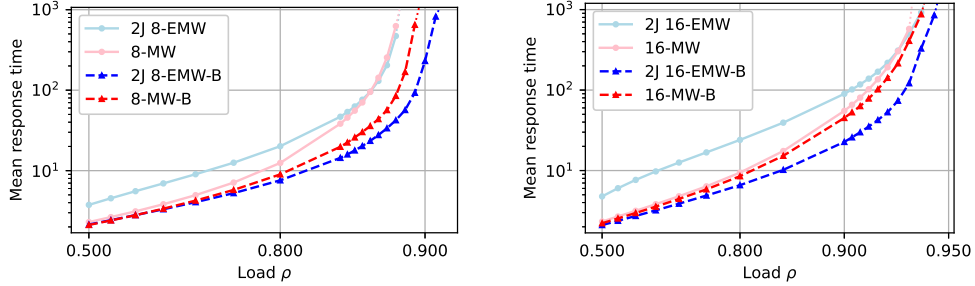
discrete symmetric resource distribution setting in which complementary job types have equal arrival rates.

Finally, we compare the performance of our theoretically-motivated policies using the efficient set E_K^{2J} and the full schedulable set C_K , to empirically show the improvement on mean response time when we choose an appropriate efficient set. According to Remark 1, K -MW is stable when the stability condition for 2J K -EMW is satisfied, and therefore, the K -MW family is also throughput-optimal when $V \sim \mathcal{TN}(0.5, 1)$. Hence, in Figs. 4a and 4b, we fix $K = 8$ and $K = 16$, respectively, and compare the mean response times under K -MW, 2J K -EMW, K -MW-B and 2J K -EMW-B. We aim to examine, when stability is guaranteed, whether considering a larger set of service options can further reduce the mean response time.

Figs. 4a and 4b show that for a fixed K , when the load is low, K -MW attains a better mean response time than 2J K -EMW. However, when the load is close to 1, all policies have similar mean response times, and their empirically-stable regions are almost identical. This behavior is possibly because, in the low load case, by considering a larger service option set, we reduce the frequency that some availability is wasted, while when the load is high, there are always many jobs needing to be served, and the waste of availability becomes rare.

In contrast, when Backfilling is applied, 2J K -EMW-B achieves a better mean response time than K -MW-B. A possible explanation is that 2J K -EMW focuses on the most important subset of service options, while Backfilling avoids the failure to pair two jobs for joint service.

Therefore, if given some K , K -EMW with some efficient set is sufficient to stabilize the system, we recommend using K -EMW-B with this efficient set, but not considering a larger set of service options, in order to achieve a larger empirically stable region and a better mean response time.



(a) Mean response times v.s. load, $K = 8$. (b) Mean response times v.s. load, $K = 16$.

Fig. 4: Log-log plot of mean response times v.s. load ρ , under 2J K -EMW, K -MW, and their counterparts with Backfilling, when $V \sim \mathcal{TN}(0.5, 1)$. Left: $K = 8$; Right: $K = 16$. Simulation duration is 10^6 jobs.

6.1.3 Weakly Decreasing (Bounded Lomax and Triangle) Resource Requirement

We then study the performance of 2-Bucket K -discretized Efficient MaxWeight (2B K -EMW) and 2B K -EMW with Backfilling (2B K -EMW-B), and verify Theorem 4, which states the throughput-optimality of the 2B K -EMW family when job resource requirement distribution has decreasing density. Specifically, we consider a bounded Lomax distribution and a triangular distribution. For the bounded Lomax distribution $\text{BLomax}(2, 1)$, the shape parameter and the scale parameter are set to 2 and 1, respectively, and the support of V is $[0, 1]$. The corresponding p.d.f. is

$$f_V(v) = \frac{8}{3}(1+v)^{-3}, \quad 0 \leq v \leq 1.$$

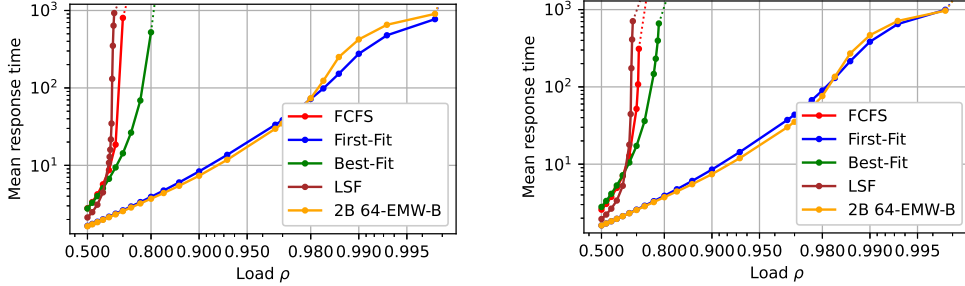
For the triangular distribution, we consider a linearly decreasing density over $[0, 1]$, given by

$$f_V(v) = 2 - 2v, \quad 0 \leq v \leq 1.$$

Both distributions have mean $\mathbb{E}V = 1/3$, and therefore, by Theorem 4, $\lambda_V^* = 3$.

In Figs. 5a and 5b, we mainly focus on comparing 2B K -EMW and 2B K -EMW-B with index-based policies. Similar to the observations in Fig. 3b, our theoretically-motivated policies achieve the largest empirically-stable region, with a boundary close to 1. The large empirically-stable region of 2B 64-EMW also verifies Theorem 4.

In addition, we observe that First-Fit performs as stably as 2B 64-EMW-B in each experiment, as in Section 6.1.2. We therefore further conjecture that, under the assumption of a weakly decreasing resource requirement density, First-Fit is throughput-optimal, extending the results of Coffman and Stolyar [4] in the centrally symmetric setting.



(a) Mean response times when $V \sim \text{BLomax}(2, 1)$. (b) Mean response times when V is triangularly distributed.

Fig. 5: Log-log plots of mean response times v.s. load ρ under theoretically-motivated policies and index-based policies when $V \sim \text{BLomax}(2, 1)$ or triangularly distributed, supported on $[0, 1]$. Simulation duration is 10^6 jobs.

6.1.4 General Lipschitz: Triangular Resource Requirement

We next consider a resource requirement distribution, the p.d.f. of which is Lipschitz continuous but neither centrally symmetric nor weakly decreasing, to empirically show that First-Fit is not always throughput-optimal, while our theoretically-motivated policies can achieve stability under higher load. We recall from Theorem 1 and Lemma 3, that the K -MW and the K -MW-B families are both throughput-optimal under this general Lipschitz setting.

We choose the symmetric triangular distribution $\text{SymmetricTri}(\frac{1}{4}, \frac{1}{2})$ as our requirement distribution, whose density is supported in $[\frac{1}{4}, \frac{1}{2}]$ and has a shape of an isosceles triangle. Formally, we say $V \sim \text{SymmetricTri}(\ell, u)$ with lower limit ℓ and upper limit u if its p.d.f. f_V satisfies that

$$f_V(v) = \begin{cases} \frac{4(v-\ell)}{(u-\ell)^2}, & \ell \leq v \leq \frac{\ell+u}{2}, \\ \frac{4(u-v)}{(u-\ell)^2}, & \frac{\ell+u}{2} < v \leq u, \\ 0, & \text{otherwise.} \end{cases}$$

Because the server can always serve at least two jobs simultaneously, $\lambda_V^* \geq 2$. According to Lemma 7, we can upper bound $\lambda_V^* \leq \frac{1}{\mathbb{E}V} = \frac{8}{3}$. However, the exact stability boundary λ_V^* is unknown, and hence we do not use the load here. We choose 30-discretized MaxWeight with backfilling (30-MW-B), because 30 has more different divisors. Empirically, K -MW and K -MW-B tend to achieve a better performance if K has more different divisors.

In addition, we evaluate the Pairwise Extreme-vertices K -Efficient MaxWeight with Backfilling policy (XP K -EMW-B), which serves as a tractable proxy for the Extreme-vertices policy (X K -EMW-B), while remaining far more efficient than full MaxWeight. By Theorem 3 and Remark 4, both the X K -EMW-B and XP K -EMW-B families are throughput-optimal. In addition, for $K = 30$, XP 30-EMW-B uses only 980 service options, whereas 30-MW-B uses 5,604 service options.

Fig. 6 compares our policies with several index-based policies. We observe that 30-MW-B and XP 30-EMW-B have indistinguishable mean response times for each arrival rate, even though XP 30-EMW-B uses more than five times fewer service options. Moreover, both our theoretically-motivated policies achieve the largest empirically-stable region and the uniformly best mean response time, while all index-based policies become unstable under high load. These simulation results demonstrate the robustness of K -MW-B and XP K -EMW-B, which consistently achieve the largest stability region across different resource requirement distribution settings. Moreover, replacing the full service-option set C_K with the smaller efficient set E_K^{XP} can substantially improve computational efficiency while preserving nearly the identical empirical performance.

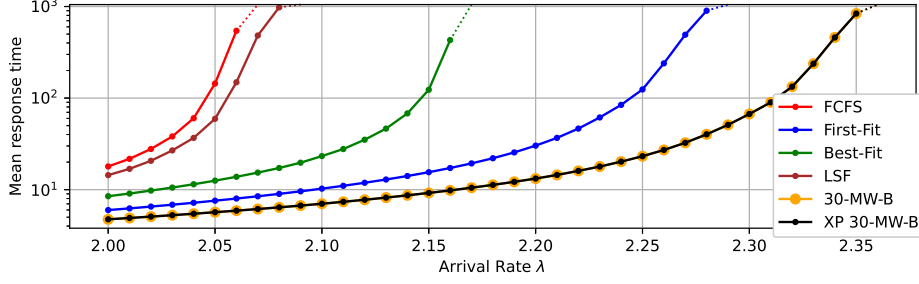


Fig. 6: The plot of mean response times under 30-MW-B, XP 30-MW-B, and index-based policies v.s. the arrival rate, where when $V \sim \text{SymmetricTri}(\frac{1}{4}, \frac{1}{2})$. Simulation duration is 10^6 jobs.

6.2 Our Theoretically-motivated Policies v.s. index-based Policies: Borg Traces

We next evaluate the practical performance of our policy using the CPU and RAM requirements from a Google Borg trace [37]², and show the state-of-the-art performance of our theoretically-motivated policies.

In our first simulation experiment, we consider an MSJ model in which the job arrival order follows the order observed in the Google Borg traces, and the resource requirement of each job is given by its rescaled CPU demand. We rescale the CPU requirements because the system capacity is unknown and the empirical distribution is highly skewed: Only a very small fraction of jobs request relatively large amounts of CPU, whereas the majority require extremely small amounts. As a result, directly using the raw data would require an excessively fine discretization, with most buckets receiving essentially zero arrivals. To address these issues, we discard the largest 10% of the observations and normalize the remaining requirements. In the second simulation experiment, we rescale the RAM requirements in the same way.

We assign each job an independent $\text{Exp}(1)$ service time. Arrival times are modeled by a Poisson process $PP(\lambda)$. However, because the resource requirements are not i.i.d. and the resulting system is non-stationary, the stability threshold λ_V^* does not apply, and hence we do not define the load in this section.

Because the two requirement distributions are approximately weakly decreasing, the 2B K -EMW-B policy is a natural fit. We set $K = 64$ and compare 2B 64-EMW-B with index-based policies. In addition, the inherently discrete value of requirements in the trace motivates us to include the class-based policy in our comparison. However, in these 1 million data points, the number of unique RAM requirements and unique CPU requirements are 3,668 and 536,229, respectively. This large number of classes makes it infeasible to run naive MaxWeight in our simulation. Instead, we propose the following class-based policy to approximate naive MaxWeight, called Pseudo MaxWeight (Pseudo-MW). Recall that naive MaxWeight performs identically to LSF

²Data available at the following repository: https://github.com/MertYILDIZ19/Google-cluster-usage-traces.v3_single_cell

when the resource requirements are continuously distributed. Similar to LSF, Pseudo-MW packs the jobs in increasing order of the ratio of resource requirement and the current number of jobs of that type. In addition to using Pseudo-MW to approximate naive MaxWeight, we further examine whether this policy achieves a similar performance to LSF.

The simulation results in Figures Figs. 7a and 7b both show that 2B 64-EMW-B significantly outperforms the index-based policies and Pseudo-MW, achieving substantially smaller mean response times. In addition, Pseudo-MW performs almost identically to LSF, which implies that when the class size is extremely large, class-based policy is not distinguishable from the index-based policies. These results show that our continuously-motivated policies can still exhibit superior performance in more realistic settings, where the resource requirements are not i.i.d., and are discrete with large support.

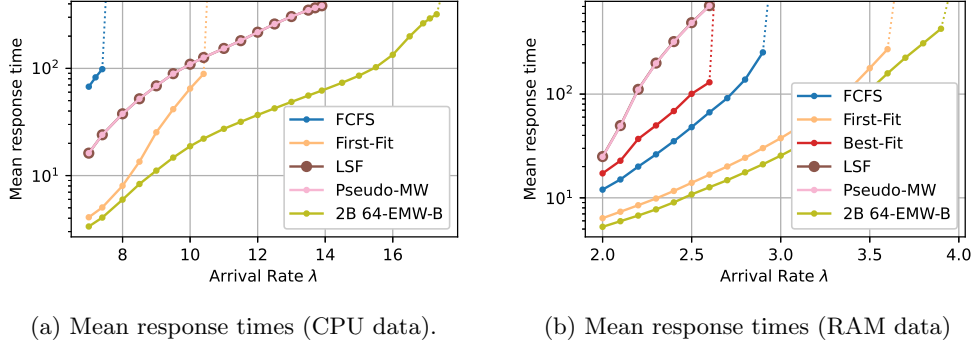


Fig. 7: The plot of mean response times under 2B 64-EMW-B, Pseudo-MW, and index-based policies v.s. the load, where job resource requirements correspond to the first one million CPU requirements and the first one million RAM requirements from the public Google Borg trace [37], respectively. Each requirement’s data is normalized by the 90% quantile of the population data, and the original data greater than the 90% quantile are excluded. Each simulation duration is the first 10^6 jobs in the trace. In the first experiment corresponding to Fig. 7a, Best-Fit is indistinguishable from unstable for any arrival rate greater than 7, and we do not include it in the plot. In both experiments, Pseudo-MW achieves almost identical mean response times to LSF, and their curves are overlapped in both Figs. 7a and 7b.

7 Conclusion

We introduce the continuous-resource multiresource-job (MRJ) system. We show the first throughput-optimality results in this setting, for the $\vec{K}$ -discretized MaxWeight family, and a related nonpreemptive policy family. Moreover, we introduce three efficient variants, which achieve the same throughput optimality with more efficient computation, under distributional assumptions.

We demonstrate via simulation the advantages of our policies when combined with Backfilling, relative to existing index-based policies. Most importantly, we evaluate our policies using Google Borg traces as job requirements. Despite the fact that these traces are not independently distributed, our simulation shows that our 2-Bucket Efficient MaxWeight achieves the best mean response time uniformly over all arrival rates.

A promising direction for future work lies in analyzing a MaxWeight variant where the discretization parameter is chosen dynamically based on the queue length, as well as nonuniform discretization. Another future direction in the single-resource setting is to prove that the throughput-optimality of First-Fit and Best-Fit, when the resource requirement distribution has symmetric or decreasing density [4].

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Declarations

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- Competing interests: The authors have no relevant financial or non-financial interests to disclose.
- Data and code availability: Our code is available at the following repository: <https://github.com/isaacg1/buckets>. For the Google Borg trace data used in our simulation, we refer the reader to the following repository: https://github.com/MertYILDIZ19/Google_cluster_usage_traces_v3_single_cell.
- Author contribution: Theoretical analysis and manuscript writing are conducted by Heyuan Yao and Izzy Grosof. Simulation software is created by Willow Kowalik and Izzy Grosof.

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Appendix A List of Notation

General Setting:

Symbol	Description
λ	Arrival rate
d	The number of resources in the MRJ setting
V	Resource requirement (joint) distribution
f_V	Probability density function of V
η^A	Arrival measure with density λf_V (9)

Multiserver (MSJ) setting under K -discretization system:

Symbol	Description
K	Discretization parameter
k	Job type, $1 \leq k \leq K$
I_k	The k -th bucket $(\frac{k-1}{K}, \frac{k}{K}]$ (5)
η_K^A	K -discretized arrival measure on $[K]$
$\Lambda_K \in \mathbb{R}^K$	Vector of arrival rates of type-1 to type- K jobs
$C_K \subset \mathbb{N}^K$	Set of schedulable service options (6)
$M \in C_K$	Service option
$M^{(k)}$	The number of type- k job(s) M serves
$E_K \subseteq C_K$	Efficient set (unspecific)
E_K^X	The Extreme-vertices efficient set
E_K^{XP}	The Pairwise Extreme-vertices efficient set (K is even)
E_K^{2B}	The 2-Bucket efficient set
E_K^{2J}	The 2-Job efficient set
β_K	A probability distribution over C_K (or some E_K)
η_K^S	K -discretized service measure on $[K]$, associated with β_K

Multiresource (MRJ) setting under $\vec{K}$ -discretization system:

Symbol	Description
$\vec{K} \in \mathbb{N}^d$	Discretization parameter
$\vec{i}$	Job type
$I_{\vec{i}}$	The $\vec{i}$ -th bucket (7)
$\Gamma_{\vec{K}}$	Job type set $\{\vec{1} \preceq \vec{i} \preceq \vec{K}\}$
$\eta_{\vec{K}}^A$	$\vec{K}$ -discretized arrival measure on $\Gamma_{\vec{K}}$ (10)
$\Lambda_{\vec{K}} \in \mathbb{R}^{\Gamma_{\vec{K}}}$	Vector of arrival rates of each type of job
$C_{\vec{K}} \subset \mathbb{N}^{\Gamma_{\vec{K}}}$	Set of schedulable service options (8)
$M \in C_{\vec{K}}$	Service option
$M^{(\vec{i})}$	The number of type- $\vec{i}$ job(s) M serves
$E_{\vec{K}} \subseteq C_{\vec{K}}$	Efficient set (unspecific)
$E_{\vec{K}}^X$	The Extreme-vertices efficient set
$E_{\vec{K}}^{2J}$	The 2-Job efficient set
$\beta_{\vec{K}}$	A probability distribution over $C_{\vec{K}}$ (or some $E_{\vec{K}}$)
$\eta_{\vec{K}}^S$	$\vec{K}$ -discretized service measure on $\Gamma_{\vec{K}}$, associated with $\beta_{\vec{K}}$ (11)

Appendix B Definitions of Some Index-Based Policies

In this section, we define First-Fit, Best-Fit, and Least-Server-First (LSF), which are the index-based policies used in Section 6 as simulation benchmarks for our policies.

We first describe First-Fit in the MSJ setting. Whenever the system state changes, First-Fit scans the jobs in the queue according to their arrival order and greedily constructs a feasible set of jobs for joint service. To be specific, suppose that the

policy has scanned the first i jobs in this order, and let x denote the total resource requirement of the jobs that have already been selected for service. Because the server has total capacity 1, the remaining available capacity is $1 - x$. If the $i + 1$ -th scanned job has a requirement $r \leq 1 - x$, then this job is selected for service, the total requirement is updated to $x + r$, and the policy moves to scan the next job. Otherwise, the job is skipped, and the policy proceeds to scan the next job without changing the selected set. First-Fit continues this procedure until all jobs in the queue have been scanned. In addition, First-Fit can be naturally extended to the MRJ setting. In this case, each job has a resource-requirement vector, and the server has unit capacity for each resource type. A scanned job is selected for service whenever its requirement for every resource type does not exceed the corresponding remaining capacity.

Likewise, in the MSJ setting, Best-Fit and LSF follow the same greedy packing rule but use different scanning orders. Best-Fit attempts to pack larger jobs first, while LSF prioritizes smaller jobs. Formally speaking, Best-Fit scans the jobs in decreasing order of their resource requirements, whereas LSF scans the jobs in increasing order of their resource requirements. We note that LSF does not need to scan the entire queue. Once the remaining capacity is smaller than the requirement of the current scanned job, any subsequently scanned job cannot fit into the remaining capacity. Hence, LSF can terminate the scan at the first time when a job fails to fit.

Appendix C Upper bound for the Supremum of Lipschitz Density Function

In this section, we derive an upper bound $L(C, d)$ on the probability density function f on $[0, 1]^d$ where we assume the Lipschitz constant of f is C . Formally speaking, we bound

$$\sup \left\{ \|f\|_\infty : f : [0, 1]^d \rightarrow \mathbb{R}_+, \|f\|_{Lip} \leq C, \int_{[0, 1]^d} f = 1 \right\}.$$

Lemma 8 (Supremum Bound of Lipschitz Continuous p.d.f. on Unit Cube) *For any probability density function f defined on $[0, 1]^d$, if f is Lipschitz continuous with Lipschitz constant C , then $\|f\|_\infty \leq L(C, d)$, where*

$$L(C, d) := \left[\frac{(d+1)\pi^{\frac{d}{2}}(\max\{C, C^*\})^d}{2^d \Gamma(\frac{d}{2} + 1)} \right]^{\frac{1}{d+1}}, \quad \text{where } C^* = \frac{(d+1)\pi^{\frac{d}{2}}}{2^d \Gamma(\frac{d}{2} + 1)}. \quad (C1)$$

Proof For any dimension d , when C is large enough, the upper bound of $\sup \|f\|_\infty$ is achieved by a function with maximum at $\vec{0}$ and directional derivative $-C$ in all directions.

In the single-resource setting, the integral of f is the area of a right triangle with side lengths L and L/C , i.e., $\frac{L^2}{2C} = 1 \Rightarrow L = (2C)^{\frac{1}{2}}$. Thus $L(C, 1) \leq \sqrt{2C}$, given large enough C . Specifically, the above holds when $L/C \leq 1$, i.e., $C \geq 2$. Thus $L(C, 1) \leq \max(\sqrt{2C}, 2)$.

In d -dimensional case, the integral of f is the same as 2^{-d} the volume of a $d+1$ -dimensional circular cone with radius L/C and height L , when $d \geq 2$. By applying the surface area

formula in d -dimensional space, $A(d, r) = \frac{d\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2}+1)}r^{d-1}$, where r denotes the radius and $\Gamma(\cdot)$ denotes the Gamma function, we have that

$$\begin{aligned} 1 &= \int_0^{L/C} \frac{d\pi^{\frac{d}{2}}}{2^d \Gamma(\frac{d}{2}+1)} r^{d-1} (L - Cr) dr = \frac{d\pi^{\frac{d}{2}}}{2^d \Gamma(\frac{d}{2}+1)} \frac{L^{d+1}}{d(d+1)C^d} \\ &\Leftrightarrow L = \left[\frac{(d+1)\pi^{\frac{d}{2}} C^d}{2^d \Gamma(\frac{d}{2}+1)} \right]^{\frac{1}{d+1}}. \end{aligned}$$

Note that therefore the discussion above requires $C \geq \frac{(d+1)\pi^{\frac{d}{2}}}{2^d \Gamma(\frac{d}{2}+1)} =: C^*$. Note that $L(C^*, d) = C^*$.

We conclude that $L(C, d)$ can be chosen as $\left[\frac{(d+1)\pi^{\frac{d}{2}} C^d}{2^d \Gamma(\frac{d}{2}+1)} \right]^{\frac{1}{d+1}}$, when $C \geq C^*$. When $C < C^*$, we can trivially choose the upper bound $L(C, d) = L(C^*, d) = C^*$. $\square$